

**BRANCH – ECE, EE, CSE (6<sup>TH</sup> SEM)**

**HSHS 3003 PROFESSIONAL ETHICS (3-0-0)**

**Course Objectives:**

This course aims to develop students understanding of ethical principles, moral reasoning and professional responsibilities. It introduces ethical theories, value systems, and the role of ethics in engineering and global contexts. The course prepares students to identify, analyze and resolve ethical dilemmas in professional and societal scenarios with integrity.

**MODULE-I (6 Hours)**

Introduction to Ethics: Basic terms- Moral, Ethics, Ethical dilemma, Emotional Intelligence; Moral development theories of Kohlberg and Piaget, View on ethics by Aristotle, Governing factors of an individual's value system, Personal and Professional Ethics

**MODULE-II (6 Hours)**

Profession and Professionalism:

Clarification of the concepts: Profession, Professional, Professionalism, Professional accountability, Professional risks, Profession and Craftsmanship, Conflict of interest, Distinguishing features of a Professional, Role and Responsibilities of Professionals, Professionals duties towards the organization and vice-a-versa.

Ethical Theories: Various ethical theories and their application – Consequentialism, Deontology, Virtue theory, Rights theory, Casuist theory

Ethical terms: Moral absolutism, Moral Relativism, Moral Pluralism etc.; Resolving Ethical Dilemma

**MODULE-III (6 Hours)**

Ethics in Engineering: Purpose and Concept of Engineering Ethics, Engineering As Social Experimentation, Types of Inquiry, Issues in Engineering Ethics

Engineers' Responsibility and Safety: Safety, Risk, Understanding the Risk, Over-estimating the Risk, Risk-Benefit Analysis, Causes of an accident and identification of the preventive measures to be taken, CASE STUDIES

**MODULE-IV (6 Hours)**

Global Ethical Issues: Different Ethical Issues in Business, Environment, IT, Bioethics, Intellectual Property Rights (IPR), Research, Media, CSR, etc.

Ethical Codes: Meaning and the Significance of Ethical Codes, The Limitations of Ethical Codes

**Course Outcome**

CO1: Define key ethical terms

CO2: Identify factors influencing value systems

CO3: Describe professional concepts

CO4: Explain the purpose of engineering ethics

**BOOKS FOR REFERENCE:**

1. R. Subramanian, "Professional Ethics", Oxford University Press
2. Mike W. Martin and Roland Schinzinger, "Ethics in Engineering", Tata McGraw Hill
3. Charles E Harris, Michael S Pritchard and Michael J Rabins, "Engineering Ethics – Concepts and Cases", Thompson Learning
4. Daniel Albuquerque, "Business Ethics", Oxford University Press
5. Edmund G. Seebauer and Robert L. Barry, "Fundamentals of Ethics", Oxford University Press

## **MODULE-1**

### **Introduction to Ethics:**

#### ***Basic Terms***

**Moral** - Morals are the prevailing standards of behavior that enable people to live cooperatively in groups. Moral refers to what societies sanction as right and acceptable.

Most people tend to act morally and follow societal guidelines. Morality often requires that people sacrifice their own short-term interests for the benefit of society. People or entities that are indifferent to right and wrong are considered amoral; while those who do evil acts are considered immoral.

While some moral principles seem to transcend time and culture, such as fairness, generally speaking, morality is not fixed. Morality describes the particular values of a specific group at a specific point in time. Historically, morality has been closely connected to religious traditions, but today its significance is equally important to the secular world. For example, businesses and government agencies have codes of ethics those employees are expected to follow.

Some philosophers make a distinction between morals and ethics. But many people use the terms morals and ethics interchangeably when talking about personal beliefs, actions, or principles. For example, it's common to say, "My morals prevent me from cheating." It's also common to use ethics in this sentence instead.

So, morals are the principles that guide individual conduct within society. And, while morals may change over time, they remain the standards of behavior that we use to judge right and wrong.

**Ethics** - Ethics is the branch of philosophy that studies moral principles, defining what's right and wrong in human conduct, acting as a system of guidelines for behavior that helps individuals and societies decide how to live good lives, make fair decisions, and treat others with respect, rooted in concepts like character, duty, and consequences. It involves systematic reflection on values, rights, responsibilities, and the language of good and bad, influencing professional conduct and societal norms. Ethics is concerned with what is good for individuals and society and is also described as moral philosophy. The term is derived from the Greek word *ethos* which can mean custom, habit, character or disposition. Ethics covers the following dilemmas:

- How to live a good life
- Our rights and responsibilities
- The language of right and wrong
- Moral decisions - what is good and bad?

**Ethical Dilemma** - An ethical dilemma is a situation in which a person faces a choice between two or more actions, each of which has a significant ethical dimension, and where choosing any one action means compromising on another ethical principle. It involves a conflict between moral imperatives, where to obey one would result in transgressing another. Ethical dilemmas are challenging because they often require a person to prioritize ethical values in a way that may not be immediately clear, and the decision can have significant consequences.

#### **Conditions for an Ethical Dilemma:**

- 1) **Presence of Conflicting Moral Principles:** There must be at least two conflicting ethical principles or values that are relevant to the situation.

**Example:** A doctor treating a teenager discovers they are using illegal drugs. The doctor faces a conflict between respecting patient confidentiality and the need to inform the teenager's parents to protect the teenager's health.

- 2) **Mutually Exclusive Choices:** The choices available to the person facing the dilemma are mutually exclusive, meaning that choosing one option necessarily precludes the other(s).

**Example:** A journalist has information that could prevent a public figure from being wrongfully accused but publishing the information would violate a promise of anonymity to a source. The journalist must choose between protecting the source and preventing an injustice.

- 3) **Significant Consequences:** The choices involved in the dilemma must have significant moral consequences.

**Example:** A software engineer discovers vulnerability in their company's software that could be exploited. Reporting it could delay the product launch, affecting the company's revenue and possibly leading to layoffs. Not reporting it could endanger users. The engineer's decision has significant consequences either way.

- 4) **Personal Responsibility for Decision:** The individual facing the dilemma must have a personal stake in the decision-making process.

**Example:** A software engineer discovers vulnerability in their company's software that could be exploited. Reporting it could delay the product launch, affecting the company's revenue and possibly leading to layoffs. Not reporting it could endanger users. The engineer's decision has significant consequences either way.

- 5) **No Clear Right or Wrong Answer:** In a true ethical dilemma, there is no clear "right" or "wrong" answer that can be easily determined by applying ethical principles.

**Example:** A manager knows that one of their employees is going through a tough personal time, which is affecting their work performance. The manager must decide whether to recommend the employee for a layoff, which is imminent due to company downsizing, or to protect the employee's job, potentially risking their own position for not meeting departmental performance criteria.

- 6) **Moral Reasoning Required:** The resolution of an ethical dilemma requires moral reasoning.

**Example:** A principal at a school learns that a popular teacher has falsified some credentials but is also aware that this teacher has had a profoundly positive impact on students' lives and the school community. The principal must engage in moral reasoning to decide whether to dismiss the teacher for the falsification or to find a way to keep the teacher on staff because of the positive contributions.

### *Emotional Intelligence*

Emotional Intelligence (EI or EQ) is the ability to understand and manage your own emotions, and to recognize and influence the emotions of others, helping you navigate social situations, build relationships, and make effective decisions. It involves skills like self-awareness, self-management, social awareness (empathy), and relationship management, and is crucial for

personal well-being and professional success, often considered as important as cognitive intelligence (IQ).

### Core Components of EI

- Self-Awareness: Recognizing your own feelings, strengths, and weaknesses, and understanding how emotions affect your thoughts and behavior.
- Self-Management (or Regulation): Controlling impulsive feelings and behaviors, adapting to changing circumstances, and staying optimistic.
- Social Awareness (Empathy): Sensing others' emotions and perspectives, and understanding social dynamics.
- Relationship Management (Social Skills): Influencing others, communicating clearly, managing conflict, and building strong bonds.

### *Moral Development Theories of Kohlberg*

Kohlberg's theory of moral development outlines how individuals progress through six stages of moral reasoning, grouped into three levels: pre-conventional, conventional, and post-conventional. At each level, people make moral decisions based on different factors, such as avoiding punishment, following laws, or following universal ethical principles. This theory shows how moral understanding evolves with age and experience.



### **Level 1 - Pre-conventional Morality**

At the pre-conventional level (most nine-year-old and younger, some over nine), we don't have a personal code of morality. Instead, our moral code is shaped by the standards of adults and the consequences of following or breaking their rules. Authority is outside the individual and reasoning is based on the physical consequences of actions.

#### Stage 1: Obedience and Punishment Orientation

The child/individual is good in order to avoid being punished. If a person is punished, they must have done wrong.

#### Stage 2: Individualism and Exchange

At this stage, children recognize that there is not just one right view that is handed down by the authorities. Different individuals have different viewpoints.

### **Level 2 - Conventional Morality**

At the conventional level (most adolescents and adults), we begin to internalize the moral standards of valued adult role models. Authority is internalized but not questioned, and reasoning is based on the norms of the group to which the person belongs.

#### Stage 3: Good Interpersonal Relationships

The child/individual is good in order to be seen as being a good person by others. Therefore, answers relate to the approval of others.

#### Stage 4: Maintaining the Social Order

The child/individual becomes aware of the wider rules of society, so judgments concern obeying the rules in order to uphold the law and to avoid guilt.

### **Level 3 - Post-conventional Morality**

Individual judgment is based on self-chosen principles, and moral reasoning is based on individual rights and justice. According to Kohlberg this level of moral reasoning is as far as most people get.

#### Stage 5: Social Contract and Individual Rights

The child/individual becomes aware that while rules/laws might exist for the good of the greatest number, there are times when they will work against the interest of particular individuals. The issues are not always clear-cut. For example, in Heinz's dilemma, the protection of life is more important than breaking the law against stealing.

#### Stage 6: Universal Principles

People at this stage have developed their own set of moral guidelines which may or may not fit the law. The principles apply to everyone. E.g., human rights, justice, and equality. The person will be prepared to act to defend these principles even if it means going against the rest of society in the process and having to pay the consequences of disapproval and or imprisonment. Kohlberg doubted few people reached this stage.

### *Moral Development Theories of Piaget*

Jean Piaget's theory of moral development is a significant framework within child pedagogy, highlighting how children's understanding of morality evolves. He identified two main stages: heteronomous morality, where young children view rules as unchangeable and focus on consequences over intentions; and autonomous morality, where older children recognize rules as socially agreed-upon and prioritize intentions, fairness and reciprocity. Educators can utilize Piaget's theory to tailor their teaching approaches, emphasizing clear rules and explanations for younger children and fostering discussions around ethical considerations, intentions and fairness as children mature.

The term "moral" is derived from the Latin word "mores" which means manners, customs and folkways. Moral development refers to the development of moral behavior and moral concepts. Moral behavior is a socially desired behavior. Moral concepts start developing when the child learns what is good and what is bad, what is right and what is wrong. Piaget (1932) used the interview method – to find out the various stages of moral development of the child. According to him, there are 4 stages of moral development which are as follows:

- 1. Anomy (Birth to years)**
- 2. Heteronomy – Authority (5 – 8 years)**
- 3. Heteronomy Reciprocity (8 – 13 years)**
- 4. Autonomy Adolescence (13 – 18 years)**

- 1. Anomy (Birth to years)** Piaget called the first stage anomy, the stage without the law. At this stage the behavior of child is neither moral nor immoral but is non-moral or moral. So, the child's behavior is not guided by moral standards. The regulators of behavior are pain and pleasure.
- 2. Heteronomy – Authority (5 – 8 years) (Morality of Constraint)** This stage of moral development may be called the discipline of artificial consequences imposed by adults. Moral development controlled by external authority. Rewards and punishment regulate moral development.
  - Children in this stage view rules and authority as fixed and unchangeable. They believe that rules are determined by adults or authorities and cannot be altered.
  - They also believe that consequences are more important than intentions. If a rule is broken, they focus on the outcome rather than the intention behind the action.
  - They have a sense of moral realism, where they believe that rules are absolute and universally applicable.
- 3. Heteronomy – Reciprocity (8 – 13 years)** At this stage, there is the morality of cooperation with peers or equals. This stage is regulated by reciprocity which implies that conformity with the group becomes necessary at this stage.
- 4. Autonomy – Adolescence (13 – 18 years)** Piaget calls the stage as equity stage. Children of this stage demand equality in moral action. The individual at this stage is fully responsible for his behavior.
  - In this stage, children begin to develop a more mature understanding of morality.
  - They recognize that rules can be created and changed through agreement among people. Rules are seen as social agreements rather than rigid authorities.

- Children become more concerned with intentions behind actions rather than just the consequences. They start to consider motives and circumstances when evaluating the morality of an action.
- They develop a sense of moral relativism, understanding that rules can vary across cultures and situations.
- Fairness and reciprocity become important concepts. They start to apply principles of equality and mutual benefit in their judgments of right and wrong.

### *View on ethics by Aristotle*

Aristotle's views on ethics are rooted in his concept of virtue ethics, which focuses on the development of good character and the cultivation of virtues as the path to living a good and meaningful life. Here are the core ideas of Aristotle's ethical theory:

#### **1. The Goal of Life: Eudaimonia (Flourishing or Well-Being)**

For Aristotle, the ultimate goal of human life is eudaimonia, which is often translated as “happiness” but more accurately means flourishing or living well. This is not a temporary state of pleasure but a lifelong process of realizing one's potential and living in accordance with reason. Eudaimonia involves living in a way that fulfills human nature, which, for Aristotle, is rational and social. It is a state where a person functions well in all areas of life: intellectually, morally, socially, and emotionally.

#### **2. Virtue as a Mean**

Aristotle argues that virtues are the qualities that enable us to achieve eudaimonia. Virtue, for him, is about finding the Golden Mean—a balance between two extremes. Each virtue lies between two vices: one of excess and one of deficiency. For example:

- **Courage** is the mean between **recklessness** (excess) and **cowardice** (deficiency).
- **Generosity** is the mean between **prodigality** (excessive generosity) and **stinginess** (deficiency).

Virtue, then, is not about following rules or adhering to rigid laws; it's about developing good habits and finding a balanced approach to life that reflects wisdom.

#### **3. Practical Wisdom (Phronesis)**

To practice virtue, Aristotle believes one needs practical wisdom (phronesis), which is the ability to make the right decisions in complex situations. It does not just about know what is good, but also about knowing how to act appropriately in different circumstances. Practical wisdom helps a person discern the right course of action and navigate the complexities of life.

#### **4. Role of Reason**

For Aristotle, reason is what distinguishes humans from other animals. Therefore, living according to reason is central to living a good life. **Rationality** is key in guiding behavior, helping individuals make choices that lead to eudaimonia.

#### **5. Moral and Intellectual Virtues**

Aristotle divides virtues into two categories:

- **Moral virtues** (like courage, temperance, and generosity) are developed through habit and practice.



- **Intellectual virtues** (like wisdom, understanding, and knowledge) are cultivated through teaching and learning.

Both types of virtues are necessary for achieving eudaimonia. Moral virtues help us act in the right way, while intellectual virtues guide us in understanding the world.

## 6. The Role of Community

Aristotle emphasizes that humans are social animals, and living ethically is always done in relation to others. The community plays a crucial role in shaping the moral character of individuals, and a good society is one that nurtures the development of virtues. Aristotle believed that we can't fully flourish in isolation. **Friendship** and **justice** are critical for a flourishing life, and the state (or political community) has an essential role in promoting the common good and creating conditions where individuals can thrive.

## 7. Ethics and Politics

For Aristotle, ethics and politics are closely connected. He argued that the goal of politics should be to create a society that helps its citizens achieve eudaimonia. Good political institutions and laws should support the development of virtue and enable people to live well. In his **Nicomachean Ethics**, Aristotle ties ethics directly to the idea of the best political system, which he believes is one that promotes the common good and fosters virtue among its citizens.

## 8. The Doctrine of the Mean and Practical Applications

Aristotle's practical advice is based on moderation. He argues that we should not seek extreme behaviors, but rather aim for moderation in all things. This is reflected in his "doctrine of the mean" where he suggests that one should always seek a balanced approach to emotions, desires, and actions.

### Criticism of Aristotle's Ethics:

- **Cultural and Gender Bias:** Some critics argue that Aristotle's ethics were shaped by the social and cultural biases of his time, especially his views on women and slaves, which are seen as inconsistent with modern ethical views.
- **Relativity of the Mean:** The concept of the "mean" may seem overly subjective. What constitutes the mean for one person may not be the same for another, depending on context and individual circumstances.
- **Eudaimonia as a Universal Goal:** Some argue that eudaimonia as a concept may not apply to everyone in the same way, especially across different cultures or individuals with varying life goals.

In sum, Aristotle's ethical theory emphasizes the development of virtuous character and the pursuit of a balanced, flourishing life. It is an ongoing process of cultivating virtues and making wise choices to achieve a harmonious and meaningful existence.

### *Governing factors of an individual's value system*

An individual's **value** system—the set of beliefs and principles that guide behavior and judgments—is shaped by multiple interrelated factors. The major governing factors are:

### **1. Family and Upbringing**

Family is the primary influence on values. Parents and elders transmit beliefs about right and wrong, discipline, honesty, respect, religion, and social behavior. Early childhood experiences strongly shape long-term values.

### **2. Culture and Tradition**

Cultural norms, customs, language, and traditions influence how individuals perceive morality, authority, relationships, and responsibilities. Cultural background plays a major role in shaping social and ethical values.

### **3. Religion and Spiritual Beliefs**

Religious teachings often provide moral guidelines related to duty, compassion, justice, honesty, and self-control. Even non-religious individuals may absorb values influenced by religious traditions in their society.

### **4. Education**

Formal education develops critical thinking, ethical reasoning, discipline, and social responsibility. Schools, teachers, and educational institutions expose individuals to moral ideals such as equality, tolerance, and respect for law.

### **5. Social Environment and Peer Group**

Friends, colleagues, and social circles influence attitudes, behavior, and priorities. Peer pressure can reinforce or challenge existing values, especially during adolescence and early adulthood.

### **6. Socio-Economic Conditions**

Economic background and life conditions affect values such as ambition, material success, frugality, empathy, and social responsibility. Struggles or privilege can shape perspectives on fairness and opportunity.

### **7. Media and Technology**

Mass media, social media, films, news, and digital platforms influence beliefs, aspirations, lifestyle choices, and ethical viewpoints. Repeated exposure can normalize certain values and behaviors.

### **8. Personal Experiences**

Life experiences such as success, failure, trauma, travel, relationships, and career challenges significantly shape an individual's value system. Personal reflection on these experiences often leads to value evolution.

### **9. Role Models and Influential Figures**

Teachers, leaders, public figures, mentors, and historical personalities influence values through their actions, ideas, and moral conduct.

### **10. Law and Political System**

Legal frameworks and political ideologies shape values related to justice, rights, equality, freedom, and civic responsibility.

## 11. Individual Reasoning and Reflection

Personal thinking, self-awareness, and moral reasoning allow individuals to accept, modify, or reject values learned from society. This factor enables growth and ethical maturity.

An individual's value system is not fixed; it evolves over time through interaction between personal experiences and social influences. Family, culture, education, religion, and personal reflection collectively govern the formation and development of values.

### *Personal Ethics*

Personal Ethics are the moral standards that help individuals determine between good and bad actions. These principles influence everyday choices, behaviours, and interactions with others. Unlike societal rules, Personal Ethics are internal and shaped by factors such as upbringing, culture, religion, and individual experiences.

Some of the key components that make an individual's moral direction include:

- a) Honesty:** Being truthful and sincere in actions
- b) Integrity:** Uphold moral ethics firmly
- c) Respect:** Value other's opinions and rights
- d) Empathy:** Understand and share other's emotions genuinely
- e) Unbiased:** Make fair decisions without prejudice
- f) Disciplined:** Maintain self-control
- g) Responsibility:** Fulfilling commitments reliably
- h) Trustworthiness:** Be reliable and keep promises
- i) Open-mindedness:** Embracing new ideas
- j) Adaptability:** Adjusting to changes and challenges

### **The Importance of Personal Ethics**

Personal Ethics shape one's character and behaviour. Here are some of the pointers that tell its importance:

- a) Lead Effectively:** They empower leaders to lead consistently following a transparent and respected ethical code. This boosts confidence and trust within the team, motivating others to follow the leader and make positive contributions to the organisation.
- b) Build Trust and Support:** Persistent ethical behaviour in all situations builds trust and credibility among peers, improving the dependability and respect of leaders and professionals.
- c) Provides Solid Foundation:** Having strong Personal Ethics assists individuals in dealing with difficult events by directing their decisions and behaviours in harmony with a defined moral compass.
- d) Streamline Decision-making Process:** Professionals with solid ethical principles can make decisions more swiftly and easily, in accordance with their own values in both personal and professional domains.
- e) Set Behavioural Standards:** Ethics form a benchmark for proper behaviour, leading people towards ethical choices and actions in both professional and personal spheres of life.

**f) Motivate Others:** People with strong Personal Ethics are often self-motivated and are ready to go the extra mile to accomplish their goals effectively.

### *Professional Ethics*

Professional people and those working in acknowledged professions exercise specialist knowledge and skill. How the use of this knowledge should be governed when providing a service to the public can be considered a moral issue and is termed professional ethics. Professionals are capable of making judgments, applying their skills and reaching informed decisions in situations that the general public cannot, because they have not received the relevant training. One of the earliest examples of professional ethics is probably the Hippocratic oath to which medical doctors still adhere to this day. Professional ethics is a set of standards adopted by a professional community. Professional ethics are regulated by standards, which are often referred to as codes of ethics. The code of ethics is very important because it gives us boundaries that we have to stay within in our professional careers. The one problem with the code of ethics is that we can't always have the answers black and white. Sometimes there are grey areas where the answers aren't so simple. Professional ethics are also known as Ethical Business Practices.

### **Components**

A number of professional organisations define their ethical approach as a number of discrete components. Typically these include:

#### **(a) Honesty**

“Honesty is the best policy” is a famous statement. Honesty refers to a facet of moral character and connotes positive and virtuous attributes such as integrity, truthfulness and straightforwardness along with the absence of lying, cheating or theft. Honesty is revered in many cultures and religions.

#### **(b) Integrity**

Integrity is a concept of consistency of actions, values, methods, measures, principles, expectations, and outcomes. In ethics, integrity is regarded as the honesty and truthfulness or accuracy of one's actions. Integrity can be regarded as the opposite of hypocrisy, in that integrity regards internal consistency as a virtue, and suggests that parties holding apparently conflicting values should account for the discrepancy or alter their beliefs.

#### **(c) Transparency**

Transparency, as used in science, engineering, business, the humanities and in a social context more generally, implies openness, communication, and accountability. Transparency is operating in such a way that it is easy for others to see what actions are performed. For example, a cashier making change at a point of sale by segregating a customer's large bills, counting up from the sale amount, and placing the change on the counter in such a way as to invite the customer to verify the amount of change demonstrates transparency.

#### **(d) Accountability**

In ethics and governance, accountability is answerability, blameworthiness, liability, and the expectation of account-giving. As an aspect of governance, it has been central to discussions related to problems in the public sector, non-profit and private (corporate) worlds. In leadership roles, accountability is the acknowledgment and assumption of responsibility for actions,

products, decisions, and policies including the administration, governance, and implementation within the scope of the role or employment position and encompassing the obligation to report explain and be answerable for resulting consequences.

**(e) Confidentiality**

Confidentiality is a set of rules or a promise that limits access or places restrictions on certain types of information. Confidentiality is ensuring that information is accessible only to those authorized to have access.

**(f) Objectivity**

Law is a system of rules and guidelines which are enforced through social institutions to govern behavior. Laws are made by governments, specifically by their legislatures. The formation of laws themselves may be influenced by a constitution (written or unwritten) and the rights encoded therein. The law shapes politics, economics and society in countless ways and serves as a social mediator of relations between people.

**(g) Respectfulness**

Respect gives a positive feeling of esteem or deference for a person or other entity (such as a nation or a religion), and also specific actions and conduct representative of that esteem. Respect can be a specific feeling of regard for the actual qualities of the one respected (e.g., “I have great respect for her judgment”). It can also be conduct in accord with a specific ethic of respect. Rude conduct is usually considered to indicate a lack of respect, disrespect, where as actions that honor somebody or something indicates respect. Specific ethics of respect are of fundamental importance to various cultures and therefore key to success in any organization is Respectfulness.

**(h) Obedience to the law**

Law is a system of rules and guidelines which are enforced through social institutions to govern behavior. Laws are made by governments, specifically by their legislatures and in organizations, by the Top Management. The formation of laws themselves may be influenced by a constitution (written or unwritten) and the rights encoded therein. The law shapes politics, economics and society in countless ways and serves as a social mediator of relations between people. And the employees working in an organization are bound to obey the Laws & Regulations of that organization.

**Need For Professional Ethics**

- (a)** Ethics means a code of conduct that directs an individual in dealing with others. Business Ethics is a form of the skill that examines ethical moralities and honesty or ethical problems that can arise in a business environment. It deals with matters regarding morals, principles, duties and corporate governance applicable to a company and its employees, customers, shareholders, media, suppliers, government and dealers. This is what the famous Henry Kravis had to say about professional ethics: “If you don't have integrity, you have nothing. You can't buy accountability. You can have all the money in the world, but if you are not a moral and ethical person, you really have nothing.”

- (b) Ethics are also related to the core of management practices such as human resource management, accounting information, production, sales and marketing, intellectual property knowledge and skill, international business and economic systems. In the corporate world, the organization's culture sets standards for shaping the difference between good or bad, right or wrong and fair or unfair. This quote by Albert Einstein says it all: "Relativity applies to physics, not ethics." The point being that it is possible to make profits without having to negotiate on ethics. And over and above the factor of correctness associated with ethics, an ethical business and its proprietors only serve themselves, their clients and the whole enterprise much better in the final reckoning.
- (c) Management gurus often preach on the advantage an ethical company has over their competitors. "A business is successful to the extent that it provides a product or service that contributes to happiness in all of its forms" – these famous words by Mihaly are a fitting description of this reality.
- (d) Lately, ethical issues in business have become more complicated because of the international and diversified nature of many big corporations and because of the difficulty of economic, social, global, political, legal, and administrative regulations and peculiarities. Thus companies have to decide whether to stick to constant ethical principles or to bend according to domestic standards and culture. It can be aptly summed up in the words of John D. Rockefeller: "I believe that every right implies a responsibility; every opportunity an obligation; every possession a duty."
- (e) In every company, the managers should remember that leading by example is the first and very important step in nurturing a culture of ethical conduct. Hence, the best way to encourage ethical behavior is by setting a good personal example. Teaching an employee ethics is not always effective. One can explain and define ethics to an adult, but understanding ethics does not necessarily result in ethical behavior. John Mackey once quoted that "Business social responsibility should not be coerced; it is a voluntary decision that the entrepreneurial leadership of every company must make on its own."
- (f) Thus, ethics are important not only in business but in all the other parts of life because it is an important base on which a civilized and cultured society is built. A business or society without ethics and scruples is only headed towards self-destruction.

## **MODULE-2**

### *Profession*

A profession is a type of work that requires specialized knowledge, training, and skills, usually backed by formal education and ethical standards. It's more than just a job—it's a recognized field where practitioners are trusted to apply their expertise responsibly for the benefit of society.

### **Key Features of a Profession**

- **Specialized Knowledge:** Professions are built on a body of knowledge derived from research, education, and training at a high level.
- **Formal Training:** Entry into a profession typically requires advanced study, certification, or licensing (e.g., medicine, law, and engineering).

- **Ethical Standards:** Professionals are expected to follow codes of conduct and act in the interest of others, not just for personal gain.
- **Public Recognition:** Society acknowledges professionals as experts in their field, granting them trust and authority.
- **Community of Practice:** A profession isn't just individual work—it's also the collective group of people engaged in that occupation (e.g., the teaching profession, the nursing profession).

Professions play a crucial role in society because they provide objective, specialized services—whether it's a doctor diagnosing illness, a lawyer offering legal counsel, or a teacher shaping future generations. They're trusted to act with integrity and competence, which is why professional misconduct is taken so seriously.

### *Professional*

A professional is someone who belongs to a recognized profession or demonstrates the qualities expected in such roles—expertise, responsibility, and adherence to ethical standards. It can describe both a person and their conduct in work settings.

A professional is a member of a profession, trained and qualified to perform specialized tasks. For example, a doctor, lawyer, or engineer. It refers to behavior or standards that align with professional norms—such as competence, skill, organization, and seriousness.

### **Characteristics of Being Professional**

- **Specialized Training:** Professionals usually undergo formal education and certification.
- **Ethical Standards:** They follow codes of conduct, ensuring integrity and accountability.
- **Competence & Skill:** Work is expected to be effective, accurate, and of high quality.
- **Appearance & Conduct:** Professionalism often extends to how one dresses, communicates, and interacts with others.
- **Distinction from Amateur:** Professionals are paid for their expertise, while amateurs may do similar work as a hobby.

### *Professionalism*

Professionalism in ethics means adhering to moral principles, standards, and values that guide conduct within a specific job or field, ensuring integrity, trust, and competence, going beyond just laws to uphold responsibilities to clients, employers, and society, often codified in a code of conduct. It combines high-level skill with honorable behavior, demanding honesty, accountability, confidentiality, impartiality, and a commitment to doing well and avoiding harm.

### *Professional Accountability*

Professional accountability means taking ownership of your actions, decisions, and outcomes in the workplace. It's about being reliable, transparent, and responsible for both successes and failures, ensuring trust and credibility in your career.

### **Key Elements of Professional Accountability**

- **Reliability:** Meeting deadlines, keeping commitments, and following through on tasks.
- **Transparency:** Communicating openly about progress, challenges, and mistakes.
- **Ethics:** Acting with integrity, even when no one is watching.

- **Learning from Mistakes:** Owning failures and using them as opportunities for growth.
- **Self-Management:** Setting goals, monitoring progress, and holding yourself accountable before others do.

#### How to Practice Professional Accountability

- **Set Clear Expectations:** Define roles, responsibilities, and deliverables.
- **Track Progress:** Use tools or check-ins to monitor commitments.
- **Own Mistakes Quickly:** Admit errors and propose solutions.
- **Seek Feedback:** Regularly ask for input to improve performance.
- **Model Accountability:** Leaders should demonstrate it to inspire teams.

#### Risks of Ignoring Accountability

- Missed deadlines and poor performance.
- Finger-pointing and toxic workplace culture.
- Loss of trust from peers and supervisors.
- Stalled career growth and missed opportunities.

#### *Professional Risks*

Professional risks in ethics involve potential breaches of moral standards that can harm individuals, organizations, or the public, stemming from issues like conflicts of interest, confidentiality breaches, negligence, discrimination, or pressure to cut corners, leading to legal trouble, financial loss, and damage to reputation. Key areas include managing safety (especially for engineers), ensuring transparency, handling data responsibly, preventing bias (like discrimination), and maintaining competence and integrity in the face of pressures.

#### Common Professional Ethical Risks

- **Compliance Failures:** Not adhering to industry regulations, labor laws, or corporate policies.
- **Conflicts of Interest:** Decisions influenced by personal gain rather than professional duty.
- **Lack of Transparency:** Withholding information or misrepresenting facts to stakeholders.
- **Poor Governance:** Weak oversight structures that allow unethical practices to flourish.
- **Cultural & Leadership Gaps:** Leaders failing to model ethical behavior, creating a permissive environment.
- **Data Misuse:** Mishandling confidential or personal information, especially in HR or finance.
- **Reputation Damage:** Loss of trust among clients, employees, or the public due to ethical lapses.

#### *Profession and Craftsmanship*

Profession and Craftsmanship both involve skilled work, but craftsmanship emphasizes deep, hands-on skill with materials, attention to detail, and artistic creation (like a potter or carpenter), often learned via apprenticeship, while a profession involves specialized knowledge, abstract problem-solving, and often formal credentials (like a lawyer or doctor), focusing on realizing



societal values or managing complex systems, though many roles blend both. Craftsmanship is about making things well; a profession is a vocation requiring broad expertise and often a distinct moral identity.

### **Profession**

- **Focus:** Applying abstract knowledge and systematic thinking to complex problems, often in service of societal goals.
- **Learning:** Formal education, certifications, degrees (e.g., MBA, Law School).
- **Values:** Intellectual rigor, ethical standards, abstract problem-solving, and realizing broader values (like justice or health).
- **Examples:** Lawyer, accountant, doctor, architect, software developer, leader.

### **Craftsmanship**

- **Focus:** Creating tangible, physical objects (woodwork, textiles, ceramics) with technical skill, artistry, and care.
- **Learning:** Often through mentorship, master-apprentice models, or hands-on practice.
- **Values:** Precision, material knowledge, beauty, utility, and innovation within tradition.
- **Examples:** Blacksmith, weaver, furniture maker, chef, fine artist.

### *Conflict of Interest*

A conflict of interest in professional ethics occurs when a person's personal interests could improperly influence—or appear to influence—their professional judgment, decisions, or responsibilities. It undermines trust, objectivity, and integrity in professional practice.

A conflict of interest is a situation in which someone in a position of trust, such as a lawyer, has competing professional or personal interests. Such competing interests can make it difficult to fulfill his or her duties impartially. A conflict of interest exists even if no unethical or improper act results from it.

A conflict of interest arises when what is in a person's best interest is not in the best interest of another person or organization to which that individual owes loyalty. For example, an employee may simultaneously help himself but hurt his employer by taking a bribe to purchase inferior goods for his company's use.

A conflict of interest can also exist when a person must answer to two different individuals or groups whose needs are at odds with each other. In this case, serving one individual or group will injure the other.

In business and law, having a “fiduciary responsibility” to someone is known as having a “duty of loyalty.” For example, auditors owe a duty of loyalty to investors who rely upon the financial reports that the auditors certify. But auditors are hired and paid directly by the companies whose reports they review. The duty of loyalty an auditor owes to investors can be at odds with the auditor's need to keep the company – its client – happy, as well as with the company's desire to look like a safe investment.

So, those of us who wish to be ethical people must consciously avoid situations where we benefit ourselves by being disloyal to others.

The various categories of conflict of interest are discussed below:

- (a) **Self-dealing:** For example, you work for government and use your official position to secure a contract for a private consulting company you own. Another instance is using your government position to get a summer job for your daughter.
- (b) **Accepting benefits:** Bribery is one example; substantial [non token] gifts are another. For example, you are the purchasing agent for your department and you accept a case of liquor from a major supplier.
- (c) **Influence peddling:** Here, the professional solicits benefits in exchange for using her influence to unfairly advance the interests of a particular party.
- (d) **Using your employer's property for private advantage:** This could be as blatant as stealing office supplies for home use. Or it might be a bit more subtle, say, using software which is licensed to your employer for private consulting work of your own. In the first case, the employer's permission eliminates the conflict; while in the second, it doesn't.
- (e) **Using confidential information:** While working for a private client, you learn that the client is planning to buy land in your region. You quickly rush out and buy the land in your wife's name.
- (f) **Outside employment or moonlighting:** An example would be setting up a business on the side that is in direct competition with your employer. Another case would be taking on so many outside clients that you don't have the time and energy to devote to your regular employer. In combination with influence peddling, it might be that a professional employed in the public service sells private consulting services to an individual with the assurance that they will secure benefits from government: "If you use my company, I am sure that you will pass the environmental review."
- (g) **Post-employment:** Here a dicey situation can be one in which a person who resigns from public or private employment and goes into business in the same area. For example, a former public servant sets up a practice lobbying the former department in which she was employed.

### *Distinguishing Features of a Professional*

#### **1. Professional appearance**

Professional appearance is one of the key characteristics of professionals, and it includes wearing appropriate attire to the workplace with proper hygiene and grooming. You can wear pants, dresses, formal skirts, formal shirts along with leather footwear to give you a clean and professional look. Some organizations may also allow wearing smart casuals at the workplace and you may choose to select your clothing that is acceptable in your organization.

#### **2. Reliability**

Professionals try to be as reliable and dependable as possible and always try to keep their commitments. Professionals are true to their word and do what they are committed to do. They never over-promise; rather they under-promise and over-deliver. Professionals are prompt in their response to colleagues and try to take follow-ups on their commitments in a timely fashion.

Professionals are punctual in everything they do and it is a key aspect of this professional characteristic. They know the importance of clarifying any uncertainties when dealing with clients or colleagues to ensure there are no mistaken assumptions or surprises.

### **3. Ethical Behaviour**

Professionals strive to live by principles they believe in and commit to doing the right thing always. Professionals believe in honesty, open disclosure and sincerity, which are all key characteristics of ethical behaviour. Most of the companies include a section on a commitment to ethical behaviour in their code of conduct to which all the professionals abide. Apart from this, professionals can adopt their own personal code of conduct and can make the same commitments individually as well.

### **4. Organized**

Being organized is one of the most important characteristics of a professional, which helps them keep their workspace organized and clutter-free. This enables them to find whatever they need in an easy way and saves precious time. Professionals keep their important paperwork and files in proper order and place. To avoid any unexpected delays, professionals keep all the required material ready well in advance in case of any important meeting or presentation. Professionals not only keep their physical spaces organized but also try to keep their data in computer systems organized.

### **5. Accountability**

As professionals accept credit for achieving every goal or objective, they also take accountability for everything they do and take full responsibility in case something does not work out. Professionals take responsibility for any mistakes that they might have made and strive to resolve the situation by taking all the necessary steps. They expect accountability not only from themselves but also from other colleagues as well to create a culture of mutual trust and competency.

### **6. Professional language**

Language is a key component that people take into account while making a perception about others. Professionals try to monitor every aspect of their behaviour, including the language they use and the way they talk. Professionals are conscious about minimizing the use of slang and try to avoid using inappropriate language not only in their workplaces but also in informal settings.

### **7. Separates personal and professional lives**

Most of the people who behave with professionalism understand why it is important to separate their personal lives from their professional lives. Separating your personal and professional lives increases efficiency at work and reduces stress. Sometimes, professionals may experience certain challenges in their personal lives, they try to maintain a clear demarcation between their personal and professional lives so it does not affect their work at the organization.

### **8. Positive Attitude**

Professionals try to keep and maintain a positive and can-do attitude in all situations in their workplace. This positive attitude at work can help the professionals in increasing their overall performance, which can increase the chances of achieving the desired goal or result. Being

positive at work not only helps the professionals but also affects the behaviour and performance of their colleagues and improves employee morale.

### **9. Emotional control**

It is often said that EQ (Emotional Quotient) is more important than IQ (Intelligence Quotient) in determining the success of an individual. Having emotional stability is important and professionals display this characteristic in various situations at the workplace. Professionals always try to maintain their calmness and composure in all adverse situations. Remaining calm helps in being rational, applying sound judgement and making good decisions during any adverse or challenging situations. This characteristic makes the professionals very reliable and others also look at them for any guidance during challenging moments.

### **10. Effective time management**

Professionals require being very efficient in their time-management skills. Time management essentially means that the employee knows how to allocate time to various things and prioritize in a way that leads to the maximum outcome. Good time-management skills can help you achieve challenging tasks under tight deadlines and can make a good impression on your manager and colleagues

### **11. Focused**

Professionals understand what needs to be accomplished to achieve their goals, objectives and get the desired results. They focus on whatever they do to achieve maximum productivity. Focus helps the professionals to achieve more in less time and improves their quality of work. While professionals can multi-task and do various things at once, they try to maintain their focus on one thing at a time to achieve the desired objectives.

### **12. Poised**

Professionals are self-assured and calm. They demonstrate poise and a confident state of being. Being poised refers to being in equilibrium, maintaining a straight posture, making eye contact when interacting with other colleagues and establishing a friendly and professional presence. Professionals who display this quality try to stay calm during tough situations and take decisions with a cool mind.

### **13. Respectful of others**

Professionals treat themselves and others with respect. They are respectful towards their managers, colleagues and support staff. While giving feedback, they convey it in a constructive and respectful manner. They also understand that, though humour is appropriate in the workplace, they must be respectful of others. People who show respect towards others are generally considered professionals by their colleagues.

### **14. Strong Communicator**

Strong communication skills are one of the key characteristics of a professional. Communication; not only means conveying your message to others in an effective and efficient way but also listening attentively to what others are saying and comprehend appropriately. Professionals have strong command over language and they leverage it to put their message across. Professionals engage in open and constructive communication with others to collaborate more effectively, which can help them accomplish a lot.

## **15. Possesses Soft Skills**

Soft skills are personality traits or habits that allow someone to interact with others in an effective way. Leadership, teamwork, adaptability, creativity and critical thinking are all examples of important soft skills. They help professionals to behave courteously when addressing colleagues and managers, use the right language when communicating and respect the opinions of others.

### *Roles of Professionals*

#### **1. Personal role:**

Professionals are human beings and have personal life. They play many roles in their personal lives, such as son, daughter, husband, wife, father, mother just like others. We are looking at whether professional spirit and professionalism has any influence on its personal life or not? The personal role is also important for their professional image. There are many factors which influence the personal life of a professional, these are religion, culture, social status, educational qualification, financial standard, legal constraints etc. In addition, they have to set their own moral and ethical standards to be followed.

#### **2. Social role:**

A professional is not only a member of professional association but also lives in a community and hence is also a member of a society and community. The social roles include, Welfare of the community, Problem solver of the community, etc. For these the professional must have knowledge, skills, strengths and special training to handle different roles.

#### **3. Ethical role:**

Many ethical issues are related to profession in everyday life. A professional must promote ethical behaviour among the people or community in which he/she lives.

#### **4. Professional Role:**

Professional role is what we will be dealing with one's responsibilities as a professional or in terms of different roles one plays in the practice of one's profession, these are, Leader: colleague; Subordinate: Savior:

The engineer as a savior, save the society from poverty, illiteracy, wastage, inefficiency, ill health, human (labor) dignity and lead it to prosperity, through technological development and social planning. For example, R.L. Stevenson

#### **5. Guardian:**

He guards the interests of the poor and general public. As one who is conversant with technology development, is given the authority befitting his expertise to determine what is best suited to the society. For example, Lawrence of Arabia (an engineer)

#### **6. Bureaucratic Servant:**

He serves the organization and the employers. The management of an enterprise fixes its goals and assigns the job of problem solving to the engineer, who accepts the challenge and shapes them into concrete achievements. For example, *Jamshedji Tata*

### **7. Social Servant:**

It is one who exhibits social responsibility. The engineer translates the interest and aspirations of the society into a reality, remembering that his true master is the society at large. For example, *Sir M. Viswesvarayya*

### **8. Social Enabler and Catalyst:**

One, who changes the society through technology,. The engineer must assist the management and the society to understand their needs and make informed decisions on the desirable technological development and minimize the negative effects of technology on people and their living environment. Thus, he shines as a social enabler and a catalyst for further growth. For example, *Sri Sundarlal Bahuguna*

### **9. Game Player:**

He is neither a servant nor master. An engineer is an assertive player, not a passive player who may carry out his master's voice. He plays a unique role successfully within the organization, enjoying the excitement of the profession and having the satisfaction of surging ahead in a competitive world. For example, *Narayanamurthy- Infosys*

## *Responsibilities of Professionals*

### **A. Senses:**

There are different senses of responsibility, such as:

**Characteristic Quality:** Primarily responsibility implies duty with care and efforts.

**Obligations:** These are one's moral responsibility i.e., duty to act right and in moral ways. The obligations such as honesty, fairness, and decency are incumbent on every one. In addition to this, we have role responsibilities assigned by taking up various roles, such as parents, inspectors, and employees. For example, a Safety Engineer has a responsibility to make regular inspections in a factory shops.

**General Moral Capacity:** One has the general capacity for moral agency, including the understanding and action on moral reasons.

**Liability and Accountability:** Liability and Accountability for actions. It means that one is liable (with a legal sense) to meet the obligations in better ways. The person is likely to respond legally, if necessary. Accountable means that one is willing to justify or defend the decisions, actions or means and outcomes. It could include offering a reasonable excuse or accepting the shame for not having met the end results or accepting the guilt for harming others. One is also answerable to the assessment by others on one's actions (means) or outcomes.

**Praiseworthiness/Blameworthiness:** When accountability for wrong actions or results is at issue, responsibility means blameworthy. When the right conduct or successful result is at issue, responsible is synonymous with praiseworthy.

### **B. Types:**

Different types of responsibilities exhibited in human transactions are:

## **I. Moral Responsibility:**

Moral responsibility as applied to a professional: A professional must be responsible morally, in creating internal good or good outcomes, and eliminating /minimizing un- intended side-effects, from engineering and technology. It includes:

**Obligations:** A commitment to moral actions (primary obligation to protect the safety of the human beings and respect their rights),

**Conscientious:** A comprehensive perspective to accept the duties, and diligently do the right things by putting their heart, head and hands (awareness of the experimental nature of the product/project, anticipating possible and unexpected outcomes and putting efforts to monitor them),

**Accountability:** (being accountable for the decisions, actions, and the results of product/ project including safety)

**Praiseworthy/Blameworthy:** Applied to context of doing things right/doing things wrongly, respectively.

## **II. Causal Responsibility:**

It is being a cause of some event. For example, a child playing with matches cause a house to burn. The child is causally responsible, but the parent who left the child with matches, is morally responsible.

## **III. Job Responsibility:**

It consists of assigned tasks at the place of employment and achieving the objectives.

## **IV. Legal Responsibility:**

It is the response required by law and includes legal obligations and accountability to meet them. Many of these responsibilities overlap with moral responsibility.

## **V. Social Responsibility:**

It consists of responsibilities of a professional towards the society within and beyond the organization.

### *Professionals' duties towards the organization and vice-a-versa*

Professionals' duties to their organization center on integrity, competence, and commitment, requiring them to uphold ethical standards, deliver quality work, maintain confidentiality, follow policies, communicate effectively, be accountable for actions, respect colleagues, and continuously develop their skills for the organization's benefit, all while fostering a positive, productive and lawful environment.

## **Core Duties & Responsibilities:**

- (a) Integrity & Ethics: Honesty, transparency, fairness, and loyalty in all dealings; avoiding gossip and conflicts of interest.
- (b) Competence & Development: Delivering quality work, staying updated in their field, and continuous learning to provide high-quality service.
- (c) Accountability & Responsibility: Taking ownership of tasks, decisions, and outcomes; admitting mistakes and seeking improvement.

- (d) Compliance: Following company policies, procedures, safety rules, and legal/regulatory guidelines.
- (e) Confidentiality: Protecting sensitive company and client information.
- (f) Communication: Communicating clearly, listening actively, and providing updates to ensure alignment.
- (g) Respect & Teamwork: Treating colleagues, clients, and managers with dignity, fostering a positive culture, and supporting the team.
- (h) Productivity & Reliability: Meeting deadlines, managing time effectively, and being a dependable presence.
- (i) Positive Representation: Acting professionally in appearance, language, and conduct to reflect well on the organization.

### **Why They Matter:**

These duties build trust, ensure smooth operations, boost employee morale, enhance the organization's reputation, and drive long-term success and sustainability.

### **Organization's Duties towards the Professional**

- (a) Fair Compensation & Benefits: Provide appropriate remuneration and rewards for work performed.
- (b) Safe & Respectful Environment: Foster a workplace free from discrimination, promoting dignity and diversity.
- (c) Clear Expectations & Feedback: Define roles, provide regular performance evaluations, and offer constructive feedback.
- (d) Development Opportunities: Invest in professional growth, training, and skill development.
- (e) Ethical Leadership: Model and enforce ethical standards, creating structures that support responsible conduct.
- (f) Work-Life Balance: Support employee well-being, recognizing their needs beyond work.

### **The Reciprocal Relationship**

This mutual duty creates a cycle where engaged, ethical professionals drive organizational success, and a supportive, ethical organization fosters professional growth and loyalty, all leading to sustained performance and positive impact.

### *Ethical Theories: Various ethical theories and their application – Consequentialism*

Consequentialism is a family of ethical theories judging actions by their outcomes, where the morally right choice produces the best overall results (good over bad) for everyone affected, focusing on future benefits, not past intentions. Key applications include Utilitarianism (greatest good/happiness for the greatest number) and Ethical Egoism (best for oneself). It's applied by calculating potential outcomes (happiness, survival, freedom) but faces challenges in predicting future results and measuring different types of good, sometimes leading to "ends justify the means" dilemmas.

### **Core Idea**

- Focus on Consequences: The morality of an act (right/wrong) depends entirely on its non-moral value or results, not the act itself or the agent's intentions.



- Goal: To achieve the best possible state of affairs in the world.

### Key Applications & Types

- (a) Utilitarianism: Aims for the greatest happiness (pleasure/absence of pain) for the most people; famously argued by J.S. Mill and Jeremy Bentham. Example: Donating money to charity instead of buying a new car, as it creates more overall good.
- (b) Ethical Egoism: Focuses on maximizing good consequences for the individual agent.
- (c) Act Consequentialism: Evaluates each individual action's consequences.
- (d) Rule Consequentialism: Judges actions based on the consequences of adopting general moral rules (e.g., "don't lie"), rather than individual acts.

### How it's Applied (Decision-Making)

- (a) Identify Alternatives: List possible actions.
- (b) Predict Outcomes: Forecast the non-moral value (good/bad) of the consequences for everyone affected by each action.
- (c) Calculate Net Good: Determine which action produces the greatest balance of good over evil.
- (d) Choose: Select the action with the best overall consequences, even if it seems counterintuitive (e.g., lying to save a life).

### Challenges & Criticisms

- (a) Predictability: It's hard to know all future consequences accurately.
- (b) Measurement: Difficulty in comparing and quantifying different types of good (e.g., freedom vs. happiness).
- (c) Justice/Rights: Can justify harming individuals (the "ends justify the means") for the greater good, potentially violating rights.

### *Ethical Theories: Various ethical theories and their application – Deontology*

Deontology is an ethical theory focused on duties and rules; asserting actions are inherently right or wrong based on moral obligations, not consequences, famously championed by Immanuel Kant, who proposed the Categorical Imperative (act only on universal rules) and treating people as ends, not means. Application involves following universal laws like "don't lie" (e.g., healthcare honesty) or professional duties (e.g., engineers' codes), but can conflict with outcomes, as seen in refusing to lie to a murderer, though some theories allow rule-bending in emergencies.

### Core Principles

- (a) Duty-Based: Morality stems from obligations, not results; an act is good if it fulfills a duty.
- (b) Intent Matters: The moral worth of an action depends on the intention behind it, not just the outcome.
- (c) Respect for Persons: Individuals have inherent worth and must be treated with dignity, as ends in themselves, not just tools for others.
- (d) Universality (Kantianism): Moral rules should be universal laws applicable to everyone, everywhere (Categorical Imperative).

## Application Examples

- (a) Medical Ethics: A doctor has a duty to tell a patient the truth about an error, even if it risks litigation, because lying disrespects the patient's autonomy and dignity (treating them as a means).
- (b) Engineering: An engineer must not break into a system to stop a missile launch if it violates their professional code (a duty), even if the consequence (missile launch) is catastrophic; this highlights deontology's rigidity.
- (c) Everyday Life: The duty to not steal or lie remains, regardless of whether stealing something small or lying to a murderer might lead to a “better” outcome.

## *Ethical Theories: Various ethical theories and their application – Virtue theory*

Virtue theory, rooted in Aristotle's philosophy, focuses on developing a moral character (virtues like honesty, courage, compassion) to guide ethical actions, asking “What would a virtuous person do?” rather than following strict rules (deontology) or calculating outcomes (Consequentialism). It involves cultivating good habits through practice and practical wisdom to achieve eudemonia (human flourishing). Applications include medicine (compassion), environmentalism (stewardship), and everyday life (honesty), judging actions by the agent's character and motivations, aiming for a good life.

## Core Principles

- (a) Character-Focused: Emphasizes being a good person, not just doing good deeds.
- (b) Virtues: Identifies moral excellences (e.g., bravery, justice, generosity) as keys to a good life, often as the mean between extremes (e.g., courage is between cowardice and rashness).
- (c) Practical Wisdom (Phronesis): The ability to discern the right thing to do in specific situations, learned through experience.
- (d) Eudemonia: The ultimate goal – human flourishing or living well, achieved through virtuous activity.

## Application in Scenarios

- (a) Stealing: Instead of “stealing is always wrong”, a virtue ethicist asks, “Would a truthful, just, generous person steal bread to feed their starving family?” The answer involves assessing the character of the act in context, often concluding a virtuous person wouldn't, even if it's difficult.
- (b) Healthcare: A virtuous doctor acts with compassion, respect, and benevolence, not just following a checklist but embodying care for the patient's well-being (flourishing).
- (c) Environmentalism: Cultivating virtues like respect for nature, stewardship, and intergenerational justice guides actions like conservation, focusing on the character of humanity's relationship with nature.

## Key Difference from Other Theories

- (a) Vs. Deontology (Duty-Based): Virtue ethics asks “What kind of person should I be?” while deontology asks “What is my duty/rule?”
- (b) Vs. Consequentialism (Outcome-Based): Virtue ethics asks “Is this action consistent with a virtuous character?” while Consequentialism asks “What action produces the best outcome?”

### *Ethical Theories: Various ethical theories and their application – Rights theory*

Rights theory, a core ethical framework, posits that individuals possess inherent moral entitlements (rights) that must be respected, prioritizing these fundamental claims over potential outcomes, guiding decisions by asking what actions uphold basic freedoms (like speech, life, healthcare access) and duties to protect them, often balancing individual liberties against societal needs, a process crucial in dilemmas from research ethics to social justice.

#### **Core Concepts of Rights Theory**

- (a) Inherent & Universal: Rights aren't granted by government but are intrinsic to being human (natural rights) or recognized by society, like freedom of expression, thought, or privacy.
- (b) Moral Claims: They establish moral claims, obligating others to respect them and creating duties.
- (c) Negative Rights: Right to be free from interference (e.g., right to not be harmed).
- (d) Positive Rights: Right to receive something (e.g., right to education, healthcare).

#### **Application in Ethical Dilemmas**

- (a) Healthcare (Drug Pricing): Focuses on the right to life and access to life-saving medicine, arguing for fair access, even if it means regulation, say Prepp.
- (b) Research (Data Accuracy): Emphasizes readers' right to accurate knowledge, demanding correction of errors as a fundamental duty.
- (c) Social Justice (Policy): Used to justify rights to welfare, education, and self-determination (second/third-generation rights).

#### **Comparison with Other Theories**

- (a) Vs. Utilitarianism: Rights theory protects individual claims even if violating them brings greater overall good (utility); Utilitarianism would weigh individual rights against the collective benefit.
- (b) Vs. Deontology: Both focus on duties, but rights theory centers on what rights exist, while general deontology emphasizes rules (like Kant's categorical imperative).
- (c) Vs. Virtue Ethics: Rights theory focuses on entitlements, while Virtue ethics focuses on the character (compassion, fairness) of the moral agent.

### *Ethical Theories: Various ethical theories and their application – Casuist theory*

Casuistry (or Casuist Theory) is a practical, case-based method in ethics that solves dilemmas by comparing them to familiar, clear-cut examples (paradigms) to find similar solutions, focusing less on abstract principles and more on wisdom from past experiences, common in bioethics and business ethics, but risks inconsistency and reliance on past biases. It uses analogies, like legal reasoning, to treat similar cases similarly, making it adaptable but potentially subjective.

#### **Core Concepts of Casuistry**

- (a) Case-Based Reasoning: Focuses on specific situations rather than universal rules.
- (b) Paradigm Cases: Relies on well-understood, extreme examples (paradigms) to understand complex problems.

- (c) Analogical Reasoning: Draws parallels between the current dilemma and past cases to find resolutions.
- (d) Expert Consensus: Seeks convergence from diverse stakeholders through shared understanding of cases.

### **Application in Professional Ethics**

- (a) Bioethics: Widely used to address complex issues like euthanasia or end-of-life care by examining specific patient scenarios.
- (b) Business Ethics: Helps in resolving dilemmas by comparing them to known situations involving conflicts of interest or loyalty.

### *Moral Absolutism*

Moral absolutism is the philosophy which says that there exists an absolute definition for moral conduct and the definition does not change with circumstances. Thus, actions are inherently moral or immoral, regardless of the beliefs and goals of the individual, society or culture that engages in the actions. It holds that morals are inherent in the laws of the universe, the nature of humanity, the will of God or some other fundamental source. Moral Absolutism can promote more extreme position that actions are moral or immoral even regardless of the circumstances in which they occur (e.g. lying is always be immoral, even if done to promote some other good, such as to save a life). Many religions have morally absolutist positions, and regard their system of morality as having been set by a deity, and therefore absolute, perfect and unchangeable.

Moral Absolutism is the ethical belief that there are absolute standards against which moral questions can be judged, and that certain actions are right or wrong, regardless of the context of the act. Thus, actions are inherently moral or immoral, regardless of the beliefs and goals of the individual, society or culture that engages in the actions. It holds that morals are inherent in the laws of the universe, the nature of humanity, the will of God or some other fundamental source. It is a form of deontology. It is the opposite of Moral Relativism, the position that moral propositions do not reflect objective and/or universal moral truths, but instead make claims relative to social, cultural, historical or personal circumstances. It is related to, but not the same as, Moral Realism (the position that certain acts are objectively right or wrong, independent of human opinion), and to Moral Universalism (the position that there is a universal ethic which applies to all people, regardless of culture, race, sex, religion, nationality, sexuality or other distinguishing feature).

The ancient Greek philosophers Plato and Aristotle believed in a kind of Absolutism or Universalism, opposing the Moral Relativism of the Sophists. Immanuel Kant was a prominent promoter of Moral Absolutism, and his formulation of the deontological theory of the Categorical Imperative was essentially absolutist in nature. Moral Absolutism has been favored historically largely because it makes the creation of laws and the upholding of the judicial system much simpler, and manifested itself in outdated concepts such as the Divine Right of Kings. Many religions have morally absolutist positions, and regard their system of morality as having been set by a deity, and therefore absolute, perfect and unchangeable. Many Christians regard Christian theology as teaching a hierarchy of moral absolutes known as graded absolutism, wherein the case of a conflict between two absolutes, the duty to obey the higher one (God) exempts one from the duty to the lower ones (fellow humans or, still lower, property). Divine Command Theory is an absolutist meta-ethical theory that an act is obligatory if (and

only if) it is commanded by God (William of Ockham argued that if God had commanded murder, then murder would indeed have been morally obligatory).

Sometimes, Moral Absolutism can mean the more extreme position that actions are moral or immoral even regardless of the circumstances in which they occur (e.g. lying is always be immoral, even if done to promote some other good, such as to save a life). In this form, it can be contrasted with Consequentialism (in which a morally right action is one that produces a good consequence or outcome, regardless of the intentions).

### **Criticisms of Moral Absolutism**

A primary criticism of Moral Absolutism regards how we come to know what the absolute morals are. For morals to be truly absolute they would have to have a universally unquestioned source, interpretation and authority, which critics claim, is impossibility. Another of the more obvious criticisms is the sheer diversity of moral opinions that exists between societies (and even within societies) in the world today, which suggests that there cannot be a single true morality. There will always be strong disagreements about which moral principles are correct and which are incorrect. For example, most people around the world probably accept the idea that we should treat others as we wish to be treated ourselves. But beyond that, people from different countries likely hold varying views about everything from the morality of abortion and capital punishment to nepotism and bribery. So, while Moral Absolutism declares a universal set of moral values, in reality, moral principles vary greatly among nations, cultures, and religions. A Consequentialist would argue that it cannot be right for a Moral Absolutist to be unprepared to kill one man in order to prevent the deaths of many others, although this would be a rather extreme and dogmatic example of Moral Absolutism.

### *Moral Relativism*

Moral Relativism says that nothing is intrinsically right or wrong and it depends upon a particular issue or situation to decide the morality of an act. Moral relativism is of the view that moral judgments are true or false only relative to some particular standpoint. Moral Relativism is a position that moral propositions do not reflect objective and/or universal moral truths, but instead make claims relative to social, cultural, historical or personal circumstances. Moral relativism is on the opposite end of the continuum from moral absolutism, which says that there is always one right answer to any ethical question. Indeed, those who adhere to moral relativism would say, “When in Rome, do as the Romans do.”

### *Moral Pluralism*

To understand moral pluralism, we must understand something of the North American context in which it has become such an issue. Moral pluralism is to be seen against a broad cultural backdrop, with three major constituent elements.

1. An intellectual pluralism, which finds its expression in postmodernism, which asserts that there is no objective truth. Plurality of beliefs is thus inevitable and to be encouraged. As a result, those who claim to have access to truth are to be regarded with something between suspicion and derision.

2. A religious pluralism, which takes its starting point from the undeniable observation that many different religions exist in North America, and proceeds to draw the conclusion that they are all equally valid manifestations of the same ultimate reality. Recognizing that religions have

ethical beliefs associated with them, this leads to the assertion that moral pluralism is to be tolerated, indeed, even celebrated. To do otherwise would seem to belittle important minority elements in North America culture.

3. A prevailing liberal political philosophy, which encourages plurality of beliefs and actions, and places considerable emphasis upon individual freedom of action. Moral pluralism is the inevitable consequence of any political philosophy which emphasizes toleration and individual self-determination.

Moral Pluralism is a philosophical approach that accepts the presence of multiple valid moral arguments and viewpoints, acknowledging that no single moral doctrine holds absolute truth.

### **Core Principles of Moral Pluralism**

At its foundation, moral pluralism advocates for the acceptance of various moral truths that co-exist. The key principles include:

- Acceptance of Diversity: Recognizing that differing moral views are often a consequence of cultural, historical, and social contexts.
- No Single Moral Truth: Acknowledging that no single moral framework possesses a monopoly on moral correctness.
- Interdependence of Values: Understanding that various moral values often interact, influence, and complement each other.
- Tolerance and Respect: Encouraging respectful dialogue and appreciation for different ethical perspectives.

### **Key Components of Moral Pluralism Theory**

The theory centers on several key components, which guide its understanding and application:

- Coexistence of Moral Frameworks: Multiple ethical systems can exist simultaneously, each providing valuable perspectives on moral issues.
- Contextual Ethics: Leads to the idea that morality can vary based on different circumstances and contexts.
- Respect for Diversity: Encourages appreciation for varied ethical opinions, fostering mutual respect and understanding.
- Dynamic Moral Discourse: Supports the notion that discussion and debate can lead to richer and more nuanced understandings of morality.

### **What are the strengths and weaknesses of moral pluralism?**

Moral pluralism's strengths include its recognition of diverse moral perspectives, fostering tolerance and inclusivity. It can accommodate complex situations by considering multiple ethical viewpoints. However, its weaknesses involve potential conflicts between competing values and difficulty in establishing clear decision-making criteria, potentially leading to inconsistency or relativism.

### ***Resolving Ethical Dilemma***

The biggest challenge of an ethical dilemma is that it does not offer an obvious solution that would comply with ethics al norms. Throughout the history of humanity, people have faced such dilemmas, and philosophers aimed and worked to find solutions to them.

The following approaches to solve an ethical dilemma were deduced:

- Refute the paradox (dilemma): The situation must be carefully analyzed. In some cases, the existence of the dilemma can be logically refuted.
- Value theory approach: Choose the alternative that offers the greater good or the lesser evil.
- Find alternative solutions: In some cases, the problem can be reconsidered, and new alternative solutions may arise.

### **Examples**

Some examples of ethical dilemma include:

- Taking credit for others' work
- Offering a client a worse product for your own profit
- Utilizing inside knowledge for your own profit

## **MODULE-3**

### *Concept of Engineering Ethics*

Engineering ethics is a set of basic rules for engineers to work in a morally and socially responsible way. It ensures their decisions consider not just their company's interests, but also the interests of clients and the public. The goal is to make sure this field of profession is trusted and respected by society.

The value promotes responsible behaviour among workers. This is highly crucial because building projects often involve big constructions that affect many people. Hence, professionals must understand the importance of acting morally. For example, they should only carry out tasks they are trained to and be honest when speaking with the public.

Everyone involved in the industry has to understand engineering ethics. Students, interns, or workers must think and act based on these fundamental principles. Because of this, every school or training provider should include this subject in their course programs. It ensures that graduates have not only skills and knowledge but also a strong sense of personal integrity.

### **Examples of Engineering Ethics**

Engineering ethics includes being honest and showing integrity, following rules, and having clear communication. It not only happens in big projects but in everyday work. Being moral means making decisions that protect people, preserve nature, and uphold the pride of the profession. According to The Engineering Council, there are four fundamental principles of this value.

#### **1. Honesty and Integrity**

Designers and technicians must always act with honesty and fairness. They are expected to be reliable and transparent in everything they do. These codes of conduct include keeping information private and not doing anything fraudulent. If they see something wrong or corrupt happening, they must take action quickly to stop it or report it. Furthermore, they must never accept bribes and should declare if they have any personal interests.

#### **2. Respect for life, law, the environment, and the public good**

Experts in this field must follow the law, ensure workers' and public safety, and consider what's best for everyone. In other words, they must put people's health and safety first and report any

risks they notice. Engineering ethics also emphasizes that they must protect people's data, care for nature, and use natural resources wisely. Above all, they should help build a better world now and for the future.

### **3. Accuracy and Rigour**

Field workers must carefully and correctly apply their skills and knowledge. This means that they should provide services only in areas where they are currently skilled or supervised by someone who is competent. For that reason, they must keep learning and improving their skills. Understanding and managing risks is a big part of their job. They must ensure their act is based on facts and fair judgment.

### **4. Leadership and Communication**

This part of engineering ethics explains that they should lead by using strong moral values. They must listen to society's concerns and help share the positive impacts of their profession. Clear and honest communication is crucial, especially when they speak in public. If they hear or see anything that goes against their values, they have to report and speak up respectfully.

#### *Purpose of Engineering Ethics*

##### **1. Safety:**

- (a) Consider a scenario where you're supposed to tell businesses and clients when their judgment is overruled due to risky conditions or when records don't adjust with relevant norms.
- (b) With these regulations, you can ensure that your work satisfies reliable standards and also prevents risks to people and community.
- (c) It is obvious to look for workplace safety with the business case along with ethical consideration.
- (d) Focusing on the well-being of workers and clients alongside developing efficiency and decreasing costs related with mishaps.

##### **2. Improves Quality:**

- (a) Engineering ethics is to improve the quality or nature of work specifically when you're in a leadership position. Let's consider you're a leader of an organization and the role is to manage people.
- (b) How? By focusing on relationships, adding value by enabling work, considering the bigger picture, relying on executive presence and shifting from a specialist to a generalist management style, you will need to adapt. The objective is to let specialists on your team do their best work.

##### **3. Public Opinion Matters:**

- (a) Ethics also contribute to changing public perceptions of engineering careers. A set of rules expects you to tell the truth in your public correspondences through unbiased, honest explanations liberated from private interest, duplicity or distortion.
- (b) For an engineer, authenticity is important in advertising and is essential to building trust, as your choices directly impact the health and safety of the general public.



- (c) It is essential to act ethically in public interactions in a time when communication skills are increasingly valued. This can help change people's perceptions of the engineering field and show that you care about ethical and responsible behavior.

#### **4. Protecting Company Interest:**

- (a) Engineering ethics can also assist in safeguarding your business' interests.
- (b) It discourages you from harming the professional reputations of other engineers in unethical ways, obtaining employment or advancement through unethical means and disclosing sensitive or confidential company information without explicit consent.
- (c) By sticking to these standards, you can assist with safeguarding your company's advantages and guarantee, you add to its prosperity.

#### **5. Sustainability:**

- (a) Engineering ethics encourage sustainability by requiring you to consider the long-term effects of your work on society and the environment.
- (b) Sustainability is essential to modern business because it can boost financial performance, expand growth opportunities and enhance your organization's reputation.
- (c) Consider the triple bottom line, one of the concepts that asserts businesses should go beyond financial performance and measure their social and environmental impacts, if you struggle to comprehend the context of sustainability in your role as an engineer.

#### **6. Ethics also protect engineers:**

- (a) Engineering ethics are not only applicable to safeguard the clients, employers and people. They also help in protecting individual engineers by preventing all professionals in the industry from acting unethically or illegally for their own benefit.
- (b) Engineers will not endeavor to acquire business or progress by untruthfully criticizing other engineers. They shall not attempt to injure, maliciously or falsely, directly or indirectly, the professional reputation, prospects, practice or employment of other engineers.
- (c) With these norms, organization or teams can enhance their productivity and performance. This also will help you in prevent the negative atmosphere in workplace which results in increase of employee engagement.

#### **7. Secure Organization Resources:**

- (a) Engineering ethics ensure that team members in a company and organizational leaders act in a manner that safeguards confidential information and intellectual property of your business.
- (b) Most of the time, your team's designs, inventions and writings are recognized as the property of your client or the person who owns those assets. You should recognize such possession arrangements before starting to work.
- (c) With this regulation, businesses and engineers can prevent theft or misuse of an organization's resources and safeguard its projects.

#### *Engineering as Social Experimentation*

Engineering projects can be viewed as social experiments in that or engineers conduct social experiment for the following reasons:

- (1) They are carried out in partial ignorance: First, any project is carried out in partial ignorance. There are uncertainties in the abstract model used for the design calculations; there are uncertainties in the precise characteristics of the materials purchased; there are uncertainties in the precision of materials processing and fabrication; there are uncertainties about the nature of the stresses the finished product will encounter.
- (2) Have uncertain outcomes: the final outcomes of engineering projects, like those of experiments, are generally uncertain. Often in engineering it is not even known what the possible outcomes may be effective engineering relies upon knowledge gained about products both before and after they leave the factory—knowledge needed for improving current products and creating better ones. Therefore:
- (3) Require monitoring and feedback: To monitor is to make periodic observations and tests in order to check for both successful performance and unintended side effects. But since the ultimate test of a product's efficiency, safety, cost-effectiveness, environmental impact, and aesthetic value lies in how well that product functions within society
- (4) Mandate obtaining informed consent from those affected.

—Learning from the Past: Usually engineers learn from their own earlier design and operating results, as well as from those of other engineers, but unfortunately that is not always the case. Iterative Design. Example iPhone 4...5...6...

- (a) Lack of established channels of communication
- (b) Misplaced pride in not asking for information
- (c) Embarrassment at failure or fear of litigation(lawsuit)
- (d) Plain neglect often impedes the flow of such information and lead to many repetitions of past mistakes.

#### **Few examples:**

1. The Titanic and Arctic
2. Ships colliding with bridges
3. Three Mile Island

These examples illustrate why it is not enough for engineers to rely on handbooks and computer programs without knowing the limits of the tables and algorithms underlying their favorite tools. Engineering, demands practitioners

Who remain alert and well informed at every stage of a project's history?

- Who exchange ideas freely with colleagues in related departments?
- Who learn from their own work?
- Who learn from other engineers?
- Learn from past and present experiences of others
- Contrasts with Standard Experiments (how is engineering different than standard experiments?): engineering differs in some respects from standard experimentation

#### *Types of Inquiry*

### **(1) Normative Inquires**

It seeks to identify and justify the morally-desirable norms or standards that should guide individuals and groups. It also has the theoretical goal of justifying particular moral judgments. Normative questions are about what ought to be and what is good, based on moral values. For example,

- (a) How far does the obligation of engineers to protect public safety extend in any given situation?
- (b) When, if ever, should engineers be expected to blow whistle on dangerous practices of their employers?
- (c) Whose values ought to be primary in making judgment about acceptable risks in design for a public transport system or a nuclear plant? Is it of management, senior engineers, government, voters or all of them?
- (d) When and why is the government justified in interfering with the organizations?
- (e) What are the reasons on which the engineers show their obligations to their employees or clients or the public?

### **(2) Conceptual Inquiries**

It is directed to clarify the meaning of concepts or ideas or principles that are expressed by words or by questions and statements. For example,

- (a) What is meant by safety?
- (b) How is it related to risk?
- (c) What is a bribe?
- (d) What is a profession?

When moral concepts are discussed, normative and conceptual issues are closely interconnected.

### **(3) Factual or Descriptive Inquiry**

It is aimed to obtain facts needed for understanding and resolving value issues. Researchers conduct factual inquiries using mathematical or statistical techniques. The inquiry provide important information on business realities, engineering practice, and the effectiveness of professional societies in fostering moral conduct, the procedures used in risk assessment, and psychological profiles of engineers. The facts provide not only the reasons for moral problems but also enable us to develop alternative ways of resolving moral problems.

For example,

- a. How were the benefits assessed?
- b. What are procedures followed in risk assessment?
- c. What are short-term and long-term effects of drinking water being polluted?
- d. Who conducted the tests on materials?

## *Issues in Engineering Ethics*

### **1. Safety and Risk**

Engineers must ensure that designs, products, and systems do not endanger human life. Examples: bridge collapses, faulty medical devices, unsafe chemical plants.

### **2. Honesty and Integrity**

Misrepresentation of data, falsifying test results, or hiding flaws violates ethical standards. Transparency in reporting failures and limitations is critical.

### 3. Environmental Responsibility

Engineering projects can cause pollution, resource depletion, or ecological damage. Ethical practice requires minimizing harm and promoting sustainability.

### 4. Conflicts of Interest

Engineers may face pressure from employers or clients to prioritize profit over safety. Ethical codes demand loyalty to public welfare above corporate interests.

### 5. Intellectual Property and Plagiarism

Copying designs or misusing proprietary knowledge undermines trust and innovation. Respecting patents and copyrights is part of professional ethics.

### 6. Equity and Fairness

Access to technology should be fair and inclusive. Issues include bias in AI systems, unequal distribution of infrastructure, or neglecting marginalized communities.

### 7. Micro vs. Macro Ethics

- **Micro-ethics:** Daily professional decisions (e.g., workplace honesty, safety checks).
- **Macro-ethics:** Larger societal issues (e.g., climate change, national infrastructure policies)

#### *Safety*

Safety is an essential ethical requirement in engineering design. Safety is concerned with avoiding certain classes of events that it is morally right to avoid. It always includes safety against unintended human death or injuries that occur as a result of the intended use of the designed object.

#### *Risk*

In professional ethics, risk is the potential for harm, loss, or negative consequences (physical, financial, reputational) arising from a professional's actions, decisions, or products, requiring ethical management to minimize these adverse effects, balancing safety with practicality, cost, and societal acceptance. It's often quantified as the probability of an event multiplied by the magnitude of its consequences ( $R = P \times C$ , where  $R$ = Risk,  $P$ = Probability of an event,  $C$ = Consequences)

#### *Understanding the Risk*

Understanding risk in professional ethics involves recognizing that risk is the potential for harm (financial, physical, reputational), and professionals have an ethical duty to identify, assess, and minimize these harms, balancing safety with practical constraints like cost, while ensuring transparency and respecting individuals' rights, especially concerning voluntary vs. involuntary exposure to danger. Key aspects include defining acceptable risk, managing conflicts of interest, fostering ethical cultures, and recognizing that while absolute safety is impossible, responsible risk management is a core moral obligation.

**Risk Definition:** The chance that something unwanted or harmful (bodily harm, financial loss, environmental damage) will occur, calculated by combining the probability of occurrence with the magnitude of the consequence.

**Safety vs. Risk:** Safety isn't the absence of risk, but rather the condition where risks are known, understood, and deemed acceptable to the relevant parties.

**Ethical Duty:** Professionals, particularly engineers, have a fundamental responsibility to protect the public, requiring them to prioritize safety and inform stakeholders about specific dangers.

### **Types of Risks & Ethical Considerations**

- **Voluntary vs. Involuntary Risks:** Risks taken willingly (like skiing) are viewed differently ethically than risks imposed on others (like pollution from a factory).
- **Public vs. Occupational Risks:** Risks to the general public (public risk) and risks specific to a job (job risk, like a miner's exposure) require different ethical management.
- **Ethical Breaches:** Risks of misconduct, conflicts of interest, bribery, or a lack of transparency threaten integrity and can cause significant harm.

### *Over-estimating the Risk*

In ethics, **overestimating risk** typically refers to cognitive biases that cause individuals or groups to have an inflated perception of their own ethical standards, an “illusion of control” over potential dangers, or a skewed assessment of the probability and severity of harms. This overestimation can lead to significant ethical lapses. Key aspects of overestimating risk in ethics include:

- (a) **Overconfidence Bias:** People tend to be overly confident in their own moral character and ability to act ethically, which can lead to decisions being made without proper ethical reflection. This “illusion of moral superiority” makes individuals complacent about their behavior.
- (b) **Misjudgment of Probability:** Individuals may assign excessive importance to unlikely events or underestimate risks they believe they can control (an “illusion of control”). This is often driven by emotional and intuitive “System 1” thinking, rather than deliberative reasoning.
- (c) **In-Group Favoritism:** We have a strong tendency to favor opinions and actions of those within our own group, which can lead to an underestimation of potential harm to out-group members, inadvertently causing discrimination.
- (d) **Rationalization and Moral Licensing:** After a questionable act, people often rationalize their behavior to avoid the psychological distress of “cognitive dissonance”. Additionally, performing a good deed can create a feeling of “moral licensing”, a belief that one is entitled to then act unethically.
- (e) **Lack of Awareness:** A major driver of unethical behavior is a lack of awareness of the ethical implications of decisions, especially when people get carried away by circumstances without considering the consequences of their actions.

In a professional context, such as risk management or research, overestimating one's ethical conduct or downplaying evidence gaps can undermine ethical principles and prevent full recognition of potential harms to stakeholders or the public. A more humble and self-reflective approach to decision-making is necessary to mitigate these biases.

### *Risk-Benefit Analysis*

The concept of “utilitarianism” suggests that the answer to a moral question is the particular course of action that would maximize human, and some extend it to the overall environment, well being. Adherents to the utilitarian concept accept the economical technique of cost benefit analysis also called marginal cost analysis as a useful tool in assessing risk.

In its application to the estimation of risk it is also referred to as “Risk Benefit Analysis”. The justification for the terminology is that the cost is usually measured in terms of deaths, injuries or other harms, as much as it could be expressed in terms of monetary figures. Either terminology is commonly used.

Cost benefit analysis possesses the same limitations of utilitarianism:

1. It may not be possible to anticipate all the costs and benefits associated with different design options, leading to an inconclusive result.
2. It may not always be possible to translate the risks and benefits into monetary or dollar terms.
3. Allowance for the distribution of costs and benefits may not be possible. A majority of the population may benefit at the expense of a smaller minority that would suffer.

In spite of these limitations cost benefit analysis has a recognized and legitimate place in risk assessment. It excels when no serious threats to individual rights are involved. It is a systematic and objective approach providing a meaningful way of comparing risks, benefits and costs using a common measure of dollars and cents.

Risk-benefit analysis is a decision-making tool used in professional ethics to evaluate the potential risks and benefits of a particular action or decision. It involves identifying and assessing the risks associated with a particular course of action or decision and weighing them against the potential benefits.

In professional ethics, risk-benefit analysis can be used to evaluate the ethical implications of different options and to determine the best course of action. For example, in healthcare ethics, risk-benefit analysis may be used to weigh the risks and benefits of a particular medical treatment, taking into account factors such as the patient's condition, the potential side effects of the treatment, and the likelihood of success.

In general, risk-benefit analysis can be a useful tool in professional ethics for ensuring that decisions are made in an informed and thoughtful manner, taking into account all relevant factors and considering the potential consequences of different courses of action.

### *Causes of an Accident*

Industrial accidents pose significant risks to the manufacturing industry, impacting productivity, employee well-being, and even the reputation of businesses. Understanding the causes of these accidents is crucial for implementing preventive measures and ensuring a safe work environment. In this article, we will explore the nine most common causes of industrial accidents and highlight their potential consequences. By addressing these issues, manufacturers can enhance safety practices, reduce accidents, and ultimately optimize operational efficiency.

## **1. Human Error**

Human error is a leading cause of industrial accidents. Despite technological advancements, the reliance on human intervention in industrial processes introduces the possibility of mistakes and oversights. Factors contributing to human error include fatigue, lack of concentration, inadequate training, and poor communication. Leveraging Predictive Maintenance tools can minimize the need for manual intervention, reducing human error. By implementing Condition Monitoring, companies can detect equipment issues early and prevent accidents before they happen.

Examples of human error in the workplace:

- Incorrectly operating machinery
- Misinterpreting safety procedures
- Failure to follow established protocols
- Improperly securing materials or equipment
- Negligence in handling hazardous substances

Mitigation steps:

- Implement comprehensive training programs for employees
- Promote a safety culture that encourages reporting and learning from mistakes
- Establish clear communication channels and protocols
- Encourage regular breaks to prevent fatigue-related errors
- Conduct regular performance evaluations to identify areas for improvement

## **2. Lack of Training and Awareness**

Inadequate training and awareness regarding safety procedures significantly increase the risk of industrial accidents. Employees who are not equipped with the necessary knowledge and skills to handle machinery, chemicals, or emergency situations may inadvertently cause accidents or fail to respond appropriately to potential hazards.

**Impact of inadequate awareness on safety procedures:**

- Increased likelihood of accidents and injuries
- Higher costs associated with medical treatment and compensation
- Potential damage to equipment and infrastructure
- Negative impact on employee morale and productivity
- Potential legal consequences and reputational damage

**Training programs and initiatives for accident prevention:**

- Develop comprehensive on-boarding programs for new employees
- Regularly update training materials to align with industry best practices
- Conduct refresher courses and workshops to reinforce safety knowledge
- Provide access to resources and educational materials on safety protocols
- Encourage employee participation in safety committees and initiatives

## **3. Equipment Failure and Malfunction**

Equipment failure and malfunction can lead to serious industrial accidents, jeopardizing the well-being of employees and disrupting production. Regular maintenance, inspections, and adherence to manufacturer guidelines are essential for preventing accidents caused by equipment failure.

**Causes of equipment failures and malfunctions:**

- Inadequate maintenance and inspection practices
- Component wear and tear
- Poor equipment design or manufacturing defects
- Exposure to harsh operating conditions
- Electrical or mechanical failures

**Examples of accidents caused by equipment failure:**

- Machine malfunctions leading to entrapment or crush injuries
- Electrical equipment malfunctions causing fires or electrical shocks
- Failure of safety systems resulting in chemical spills
- Malfunctioning control panels leading to incorrect settings
- Equipment breakdown causing production delays and financial losses

**Importance of regular maintenance and inspections:**

- Implement preventive maintenance schedules for all equipment
- Train employees on proper maintenance procedures and protocols
- Conduct regular inspections to identify potential issues
- Address equipment failures promptly to avoid accidents
- Collaborate with manufacturers and suppliers to ensure equipment reliability
- Regular Preventive Maintenance helps avoid critical equipment failures that can lead to industrial accidents.

#### **4. Unsafe Work Environment**

An unsafe work environment significantly increases the risk of accidents and injuries for employees. Factors contributing to an unsafe work environment include poor housekeeping practices, lack of safety measures, and inadequate protective equipment.

Poor housekeeping practices and their consequences:

- Tripping hazards due to cluttered work areas
- Accumulation of flammable materials increasing fire risks
- Blocked emergency exits impairing evacuation efforts
- Exposure to toxic substances due to improper storage
- Increased likelihood of slips, trips, and falls

Impact of inadequate safety measures and protective equipment:

- Greater risk of injuries and accidents
- Longer recovery times for injured employees
- Increased healthcare costs
- Higher workers' compensation claims
- Negative impact on employee morale and retention



## **5. Improper Use of Tools and Equipment**

The improper use of tools and equipment is a common cause of industrial accidents. Failure to follow manufacturer guidelines, inadequate training, and misuse of equipment can result in serious injuries and equipment damage.

Risks associated with improper tool and equipment use:

- Cuts, lacerations, and puncture wounds
- Crushing injuries caused by improper lifting techniques
- Electrical shocks due to incorrect handling of electrical equipment
- Equipment malfunctions resulting from misuse
- Exposure to hazardous materials due to improper handling

Importance of following manufacturer guidelines:

- Ensure employees receive proper training on equipment usage
- Clearly label tools and equipment with instructions and safety precautions
- Regularly inspect tools for damage or defects
- Promote a culture of accountability for equipment use
- Encourage reporting of any equipment-related issues or concerns

## **6. Lack of Communication and Coordination**

Effective communication and coordination among employees are vital for maintaining a safe work environment. Miscommunication, misunderstandings, and lack of coordination can lead to accidents and failures in implementing necessary safety measures.

Importance of effective communication among employees:

- Ensuring clear understanding of safety protocols and procedures
- Facilitating timely reporting of hazards and potential risks
- Enabling efficient emergency response and evacuation
- Promoting teamwork and collaboration in implementing safety measures
- Reducing the likelihood of errors and misunderstandings

Impact of poor coordination on safety measures:

- Delayed response to safety incidents or emergencies
- Inefficient allocation of resources during crises
- Increased likelihood of accidents due to miscommunication
- Lower morale and decreased employee engagement
- Compromised safety culture and compliance

Examples of accidents caused by miscommunication:

- Failure to warn colleagues about potential hazards
- Confusion regarding lockout/tag-out procedures
- Inadequate communication during equipment maintenance
- Errors in interpreting work instructions or specifications
- Lack of coordination leading to collision or contact accidents

## **7. Fatigue and Stress**

Fatigue and stress have a significant impact on employee performance and can contribute to industrial accidents. Long working hours, insufficient breaks, and demanding work environments increase the likelihood of errors, decreased concentration, and impaired decision-making.

Effects of fatigue and stress on employee performance:

- Reduced alertness and reaction time
- Impaired judgment and decision-making abilities
- Inability to concentrate and follow safety procedures
- Increased likelihood of errors and accidents
- Higher risk of physical and mental health issues

Strategies for managing fatigue and stress in the workplace:

- Implement regular shift rotations and breaks to prevent excessive fatigue
- Encourage a healthy work-life balance for employees
- Provide access to stress management programs and resources
- Train supervisors to identify signs of fatigue and stress
- Promote a supportive and positive work environment

## **8. Lack of Risk Assessment and Hazard Identification**

Conducting thorough risk assessments and identifying potential hazards are critical steps in preventing industrial accidents. Failure to recognize and mitigate workplace hazards significantly increases the likelihood of accidents and injuries.

Importance of conducting risk assessments:

- Identifying potential risks and hazards in the work environment
- Developing appropriate control measures and safety protocols
- Enhancing employee awareness and understanding of potential dangers
- Enabling proactive accident prevention rather than reactive responses
- Complying with regulatory requirements and industry standards

Failure to identify and mitigate workplace hazards:

- Increased risk of accidents, injuries, and illnesses
- Potential legal repercussions and fines
- Higher healthcare and insurance costs
- Damage to equipment and infrastructure
- Negative impact on employee morale and productivity

Examples of accidents resulting from inadequate risk assessment:

- Chemical spills due to improper storage or handling
- Slips and falls caused by unaddressed wet surfaces
- Electrical shocks due to insufficient grounding
- Exposure to hazardous substances without proper protection
- Fire hazards resulting from improper storage of flammable materials

## **9. Non-Compliance with Safety Regulations**

Non-compliance with safety regulations is a significant risk factor for industrial accidents. Failure to adhere to local and international safety standards can result in serious consequences, including injuries, legal implications, and reputational damage.

Consequences of non-compliance with safety regulations:

- Increased likelihood of accidents and injuries
- Legal penalties, fines, and litigation
- Damage to business reputation and customer trust
- Higher insurance premiums and costs
- Potential suspension or closure of operations

Importance of following local and international standards:

- Stay informed about relevant safety regulations and updates
- Establish internal safety policies and procedures
- Train employees on compliance requirements
- Conduct regular safety audits and inspections
- Implement corrective actions to address non-compliance

Industrial accidents can have severe consequences for both employees and businesses. By understanding the nine most common causes of industrial accidents and implementing appropriate preventive measures, manufacturers can significantly reduce the risks associated with workplace incidents. Prioritizing proper training and awareness, equipment maintenance, a safe work environment, effective communication, employee well-being, risk assessment, and compliance with safety regulations will contribute to a safer and more productive manufacturing industry.

### *Identification of the Preventive Measures to be taken*

#### **1. Create a safety and wellness plan**

One key step in preventing accidents at work is developing a comprehensive safety and wellness plan. This document should explain all safety policies and appropriate procedures for:

- Completing safety related tasks
- Alerting management to possible hazards
- Documenting any safety incidents

Having a plan helps employees understand exactly what management's expectations are. This can help prevent miscommunication and possible legal issues if accidents occur.

#### **2. Require pre-employment physicals for demanding positions**

Many organizations require employees in certain types of roles to undergo physical and mental screenings before taking a job. This ensures employees are physically and mentally capable of performing the duties each position demands. For example, pilots complete thorough visions and hearing tests as part of the recruitment process, and truck drivers must pass regular health screenings to continue working. Pre-employment physicals may include the following elements:

- Physical exam
- Review of medical history

- Vision and hearing tests
- Lung capacity test
- Cardiac stress test
- Drug and alcohol screening
- Pull and grip strength test

Employers may require employees in labor intensive positions to have physicals annually.

### **3. Hold regular training**

Educating team members on potential safety hazards and proper workplace procedures can help employees understand the risks associated with their jobs and how they can reduce their chances of an accident or injury. Holding regular trainings informs employees about changes to safety standards or to reinforce current policies. Depending on your industry, this training may include:

- Proper lifting techniques
- Procedures for handling hazardous material
- Accident reporting
- Locations of emergency equipment like fire extinguishers and AED machines

### **4. Identify safety concerns**

Complete a safety evaluation of your business to identify trends and possible hazards. You can:

- Walk the premises to check for potential risks
- Review accident reports from the previous year
- Hire a safety consultant to evaluate your workplace

Focusing on the specific areas related to your organization can help you customize your safety and wellness plan with policies that apply to your business and your employees.

### **5. Provide protective gear**

Depending on the nature of your work environment, you may need to provide Personal Protective Equipment (PPE) to keep employees safe. PPE can include:

- Helmets
- Goggles
- Reflective vests or other high-visibility clothing
- Gloves
- Ear coverings
- Waist belts
- Harnesses
- Foot protection

Train new employees on how to use their equipment properly. Consider performing unannounced safety audits to make sure employees are using their protective equipment appropriately. In addition to personal gear, you may need to provide team members with other safety equipment for their work zones, like traffic cones, signs and first aid kits.

### **6. Address staffing needs**

You can help prevent accidents in the workplace by meeting your organization's staffing needs. Having enough team members to complete tasks on schedule can help prevent burnout and overwhelm among employees. Hiring enough staff can reduce the need for overtime and allow employees more breaks. When employees get enough rest and time off, they can feel more motivated, more engaged and more focused, which can help prevent accidents.

### **7. Keep work spaces clean and organized**

You can prevent accidents in the workplace by maintaining a clean, organized environment. Make cleaning procedures a part of employees' regular responsibilities. Ensure work areas are free of clutter, spills and tripping hazards like loose rugs or wiring. In warehouses or other industrial settings, provide cleaning stations throughout the workplace so employees can clean up spills quickly. Clearly mark off these stations with signs or lights so that team members can find them easily.

### **8. Dress for the weather and job conditions**

Train employees to prepare for the specific conditions associated with their jobs. This includes wearing proper clothing for the weather and job duties. For example, employees working outdoors in a hot climate should wear lightweight, breathable fabrics. Those working in cold environments should wear warm layers. If a job requires lots of movement, employees need to wear clothing that is flexible and allows them to bend, crawl and reach easily.

### **9. Install proper lighting**

Make sure the workplace is well lit, especially:

- Work areas
- Footpaths
- Entrances and exits
- Outdoor spaces
- Parking areas

If employees are working offsite, you may need to provide lighting equipment, like a spotlight or headlamps, to ensure their safety.

### **10. Perform regular inspections**

To make sure your workplace is safe, perform regular inspections to identify possible hazards and ensure employees are following all safety policies. Keep a record of safety violations or concerns that you can use to modify your company's safety and wellness plan. It's also important to address safety issues with employees right away and explain the reasons for the policies. Having a record of incidents can alert you to repeat issues with employees, which may require disciplinary action. You should also inspect any company vehicles regularly and perform routine maintenance to keep the vehicles safe and operational, such as:

- Oil changes
- Tire balancing, rotation or replacement
- Windshield wiper replacement
- Brake system monitoring

### **11. Hire a safety officer**

To help your company maintain a consistent focus on safety and accident prevention, consider hiring a safety officer. A safety officer is responsible for determining risks, developing a safety plan, implementing safety procedures and enforcing policies. They can also make sure your company complies with all state and federal safety regulations for your industry.

## **MODULE-4**

### *Different Ethical Issues in Business*

#### **1. Discrimination and Harassment**

Discrimination and harassment are major ethical issues for HR professionals and managers, with serious impacts on an organization's finances and reputation. Anti-discrimination laws in many countries aim to protect employees from such unfair treatment. Some anti-discrimination areas include:

- **Age:** Organizations and internal policies cannot discriminate against employees who are older.
- **Disability:** To prevent disability discrimination, it's important to accommodate and provide equal treatment for employees with mental or physical disabilities.
- **Equal pay:** Equal pay focuses on ensuring that all employees receive equal compensation for similar work, regardless of religion, gender or race.
- **Pregnancy:** Pregnant employees have a right not to be discriminated against on account of their pregnancy.
- **Race:** All employees should receive equal treatment, regardless of race.
- **Religion:** Employees' religious beliefs should not affect how anyone within the organization treats them.
- **Sex and Gender:** An employee's sex and gender identity should not influence their treatment while working at an organization.

HR professionals and senior managers should educate employees on discrimination issues and foster a positive work culture to combat it. Clearly communicate the disciplinary consequences of discriminatory behavior to all staff. Strive to hire individuals from diverse backgrounds to promote a varied workforce. Consider age, religion, and culture when developing internal policies to better accommodate employees' needs.

#### **2. Workplace health and safety**

All employees have a right to safe working conditions and a positive working environment. Some of the most common employee safety considerations include:

- **Fall Protection:** This involves measures to protect employees against falls, such as guard rails.
- **Hazard Communication:** Identify any harmful substances employees work with and communicate how to handle these hazardous materials safely.
- **Scaffolding:** The HR department in construction or maintenance organizations is obliged to guide employees about the maximum weight numbers structures can handle.
- **Respiratory Protection:** If relevant, provide guidelines about emergency procedures and the standards applicable to the use of respiratory equipment.
- **Lockout, Tag Out:** This involves specifying the control procedures for dangerous machines and hazardous energy sources, such as gas and oil.
- **Industrial Trucks:** It's important to ensure that the required safety standards for trucks are in place to protect employees.

- **Ladders:** Before using ladders, employees must be given an understanding of the weight that the ladder can support.
- **Electrical Wiring Methods:** Create procedures for electrical and wiring tasks. For example, these guidelines can specify how employees can create a circuit to reduce electromagnetic interference.
- **Machine Guarding:** It's important to provide operation guarding instructions for items such as guillotine cutters, power presses, shears and other devices where applicable.
- **General Electrical Regulations:** Developing general electrical regulations for employees is critical for safety in work environments that require the frequent use of electrical equipment. For example, employees should never place conductors or equipment in damp or wet locations.

Health and safety guidelines should address not just physical harm but also psychosocial risks, work-related stress, and mental health issues. Factors like high work demands, job insecurity, effort-reward imbalance, and low autonomy can contribute to these health-related behavioral risks.

### **3. Whistle-blowing or Social Media Rants**

Using social media has become widespread, making employees' online conduct a critical consideration in their employment status. Punishing employees for inappropriate social media posts raises ethical concerns, as negative posts can affect how employees are treated. When an employee's social media posts result in a loss of business or give the organization a negative reputation, you may decide to fire them. Specifying inappropriate social media behaviour in company policies helps employees know what to avoid. As an HR professional or manager, you cannot penalize employees who whistle-blow to regulators or highlight workplace violations online, unless it harms the company's business.

### **4. Ethics in Accounting Practices**

Laws require organizations to maintain accurate bookkeeping practices. Unethical accounting practices are a serious issue, especially for publicly traded companies. The legislation specifies financial report requirements aimed at protecting shareholders and consumers. All organizations have to keep accurate financial records and pay taxes to attract investment and business partners regardless of the size of the company.

### **5. Corporate Espionage and Non-disclosure**

Many organizations are at risk that current and former employees may steal information, such as client data, for use by competitors. Stealing an organization's intellectual property or illegally distributing private client information constitutes corporate espionage. This is why it can be helpful to require mandatory nondisclosure agreements. As an HR professional or manager, you may also wish to set strict financial penalties for violations to discourage these types of ethical violations.

### **6. Technology and Privacy Practices**

Developments in an organization's technological security capabilities may pose privacy concerns for both employees and clients. You can monitor employees' activity on their work computers and devices provided by the company. As a manager, you can use this method of electronic surveillance to ensure productivity and efficiency if it doesn't violate the employee's privacy.

Electronic surveillance, such as monitoring internet use and keystrokes, should be conducted transparently with employees. To avoid ethical issues, it's important to clearly communicate the benefits of the surveillance system to all staff.

### **7. Nepotism or favouritism**

As a hiring manager or HR professional, you may want to employ an acquaintance or family member because of your connection to them. Even if you adhere to recruitment policies to ensure a fair process, some employees may still consider this as nepotism or favouritism. Favouritism occurs when managers treat some employees better than others for no professional reason. This can reduce productivity and job satisfaction in other employees, which may negatively impact the entire organization.

### **8. Environmental Responsibility**

Many organizations are increasing corporate social responsibility activities. You can help create policies that ensure the organization you work for acts in a responsible way towards employees, the community and the environment. If you work for a large company in the oil or farming sectors, you have a more significant corporate social responsibility because of the organization's significant impact on the environment. If you work for a smaller organization, you may wish to reduce the company's impact on air and water quality.

#### *Different Ethical Issues in Environment*

- 1) Resource consumption patterns and the need for equitable utilization:** It deals with how we utilize and distribute resources. The disparity between haves and have-nots is widening. There is a disparity between the individuals, communities and countries in usage of resources. The well-to-do, educated urban dweller consumes much larger quantities of resources and energy than the traditional rural individual. This unequal distribution of wealth and access to land and its resources is a serious environmental concern. An equitable sharing of resources forms the basis of sustainable development for urban, rural and wilderness-dwelling communities.
- 2) Equity-disparity in the northern and southern countries:** It is concerned with who owns resources and how they are distributed. People living in the economically-advanced nations use greater amounts of resources and energy per individual and also waste more resources. This is at the cost poor people who are resource-dependent and live in developing nations.
- 3) Urban-rural equity issues:** The common property of rural communities has increasingly been used to supply the needs of the urban and industrial sectors. As the rural sector supplies food and a part of the energy needs (mainly fuel wood) to most towns and cities in India, the common lands of the rural sector are being depleted of their resources.
- 4) Preserving resources for future generations:** This ethical issue must be considered when we use resources unsustainably. If we overuse and misuse resources and energy from fossil fuels then our future generations will find survival very difficult.



- 5) **The rights of animals:** The plants and animals that share the Earth with us too have a right to live and share the Earth's resources and living space. We have no right to push a species that has taken millions of years to evolve towards extinction. Cruelty to animals is a crime that must be regarded seriously and action must be taken against offenders.
- 6) **The ethical basis of environment education and awareness:** The most important concern is related to creating an ethos that will support a sustainable lifestyle in society. The Supreme Court of our country has ordered that every young individual at school and college level be exposed to a course on environment. There are two aspects that are closely connected with ethical issues that are related to our environment. These are based on valuing nature as a resource and appreciating the beauty of nature and treasuring the magnificence of the wilderness.
- 7) **The conservation ethic and traditional value systems of India:** During olden days, people have always valued mountains, rivers, forests, trees and several animals. Thus, much of nature was venerated and protected. Certain species of trees have been protected as they are valued for their fruit or flowers. Traditions held the animals/species as an important aspect of nature were the basis of local life-support systems and were integral to bring about a harmonious life.

#### *Different Ethical Issues in IT*

##### **1. Personal Privacy**

It is an important aspect of ethical issues in information technology. IT facilitates the users having their own hardware, operating system and software tools to access the servers that are connected to each other and to the users by a network. Due to the distribution of the network on a large scale, data or information transfer in a big amount takes place which leads to the hidden chances of disclosing information and violating the privacy of any individuals or a group. It is a major challenge for IT society and organizations to maintain the privacy and integrity of data. Accidental disclosure to inappropriate individuals and provisions to protect the accuracy of data also comes in the privacy issue.

##### **2. Access Right**

The second aspect of ethical issues in information technology is access right. Access right becomes a high priority issue for the IT and cyberspace with the great advancement in technology. E-commerce and Electronic payment systems evolution on the internet heightened this issue for various corporate organizations and government agencies. Network on the internet cannot be made secure from unauthorized access. Generally, the intrusion detection systems are used to determine whether the user is an intruder or an appropriate user.

##### **3. Harmful Actions**

Harmful actions in the computer ethics refers to the damage or negative consequences to the IT such as loss of important information, loss of property, loss of ownership, destruction of property and undesirable substantial impacts. This principle of ethical conduct restricts any outsiders from the use of information technology in manner which leads to any loss to any of the users, employees, employers and the general public. Typically, these actions comprise of the intentional destruction or alteration of files and program which drives a serious loss of resources.

To recover from the harmful actions extra time and efforts are required to remove the viruses from the computer systems.

#### **4. Patents**

It is more difficult to deal with these types of ethical issues. A patent can preserve the unique and secret aspect of an idea. Obtaining a patent is very difficult as compared with obtaining a copyright. A thorough disclosure is required with the software. The patent holder has to reveal the full details of a program to a proficient programmer for building a program.

#### **5. Copyright**

The information security specialists are to be familiar with necessary concept of the copyright law. Copyright law works as a very powerful legal tool in protecting computer software, both before a security breach and surely after a security breach. This type of breach could be the mishandling and misuse of data, computer programs, documentation and similar material. In many countries, copyright legislation is amended or revised to provide explicit laws to protect computer programs.

#### **6. Trade Secrets**

Trade secrets are also a significant ethical issue in information technology. A trade secret secures something of value and usefulness. This law protects the private aspects of ideas which is known only to discover or his confidants. Once disclosed, trade secret is lost as such and is only protected by the law for trade secrets. The application of trade secret law is very broad in the computer range, where even a slight head start in the advancement of software or hardware can provide a significant competitive influence.

#### **7. Liability**

One should be aware of the liability issue in making ethical decisions. Software developer makes promises and assertions to the user about the nature and quality of the product that can be restricted as an express warranty. Programmers or retailers possess the legitimate to determine the express warranties. Thus they have to be practical when they define any claims and predictions about the capacities, quality and nature of their software or hardware. Every word they say about their product may be as legally valid as stated in written. All agreements should be in writing to protect against liability. A disclaimer of express warranties can free a supplier from being held responsible of informal, speculative statements or forecasting made during the agreement stages.

#### **8. Piracy**

Piracy is an activity in which the creation of illegal copy of the software is made. It is entirely up to the owner of the software as to whether or not users can make backup copies of their software. As laws made for copyright protection are evolving, also legislation that would stop unauthorized duplication of software is in consideration. The software industry is prepared to do encounter against software piracy. The courts are dealing with an increasing number of actions concerning the protection of software.

## *Different Ethical Issues in Bioethics*

### **1. The issue with informed consent**

Conventionally, informed consent is in the form of written and signed documents. However, an important part of it is informing the patients about their rights, the purpose of the study, the procedures to be undergone, the potential risks and/or benefits of participation, and alternative treatment plans.

For the consent to be true some points should be investigated. Firstly, it is important that the person who gives consent is capable of doing so, i.e. they are competent. Secondly, the individual has to have a full understanding of the procedure or treatment they receive. Lastly, the person should have the right not to consent. Ethical considerations surrounding research involving human subjects emphasize the importance of informed consent and highlight the challenges of maintaining anonymity in data collection.

The infamous Tuskegee Syphilis Study is an example of unethical conduction of study. It was going between 1932 and 72 and it aimed to observe the untreated syphilis. The participants (600 impoverished African Americans) were promised free of charge healthcare but were never informed that they had syphilis and that it was treatable. More than 120 of the participants died from syphilis and its complications because of the study. [10] This historical context underscores the necessity for strict ethical guidelines in human subjects research.

### **2. Use of placebo**

Placebo treatment is used in randomized controlled trials and it aims to demonstrate the effectiveness of other treatments. The ethical use of placebos in clinical trials is questionable, particularly when effective treatments are already available. There is a consensus in the research community that placebo control groups should not be included for life-threatening conditions or harmful non-reversing conditions when an effective treatment exists. Giving a placebo in those cases violates informed consent and threatens the patient-doctor's trust. Another question is also raised – is it ethical to prescribe a placebo only for its placebo effect and as a possible alternative pain relief?

### **3. Privacy and confidentiality**

In this aspect the principle of autonomy and the principle of justice are applied. It is accepted that confidentiality can be breached if:

- There is a legally authorized request;
- The patient's best interest requires it;
- The society or a third party must be protected from a major harm or threat.

This includes infectious diseases, gunshot wounds, sexually-transmitted diseases, genetic risks, violence etc.

### **4. Conflicts of interest**

These give bias of the research or practice and include bias towards the interests of hospital management, towards individuals' perspectives and interests, towards moral beliefs, political bias, and others. According to criteria for good ethical argumentation, such as the Rapoport rules, declarations of biases together with (or as part of) declarations of conflicts of interest in research papers may reduce the biases or the negative effect from them.

## **5. Animal testing**

Animal testing is often seen as cruelty but historically is proven to be a mandatory part of the treatment development. If animal testing is not used in harmful ways and for one's entertainment it is a legitimate need to improve the condition of people or save their lives. A famous example of reduction of the animal testing gone bad is the thalidomide development. It was tested only in non-pregnant rodents and released on the market. It led to a huge number of birth defects all over the world and was withdrawn.

## **6. End-of-life decisions**

End-of-life decisions regard Do Not Resuscitate (DNR) requests, euthanasia, and physician-assisted suicide. An early DNR code would save the patient and family futile interventions and pain and it saves time and resources. However, it is still a breach of the rule to not harm. Referring to end-of-life decisions, ethical and medical decisions should be carefully weighted as well as therapeutic benefits against risks.

### *Intellectual Property Rights (IPR)*

Intellectual property rights are the rights given to each and every person for the creation of new things according to their minds. IPR usually give the creator a complete right over the use of his/her creation for a certain period of time.

Intellectual property rights are the legal rights that cover the benefits given to individuals who are the owners and inventors of work and have created something unique with their intellectual creativity or capability. Every person related to areas such as literature, music, invention, etc., can be granted such rights, which can then be used in their business practices by them.

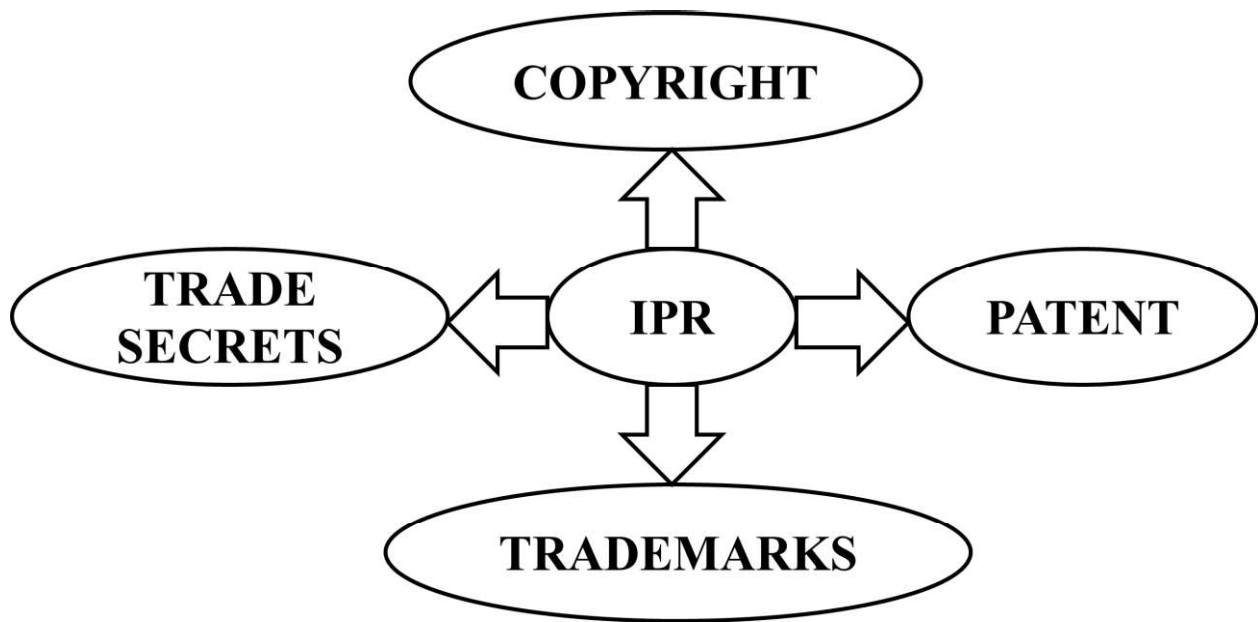
The creator/inventor gets complete rights against any misuse or use of work without his/her prior information. However, the rights are issued for a limited period of time to maintain equilibrium.

### **What are Intellectual Properties?**

- 1) Industrial designs
- 2) Scientific discoveries
- 3) Protection against unfair competition
- 4) Literary, artistic, and scientific works
- 5) Inventions in all fields of human endeavor
- 6) Trademarks, service marks, commercial names, and designations

### **Types of Intellectual Property Rights:**

Intellectual Property Rights can be classified into four types:



1. **Copyright:** Copyright is a term that describes ownership or control of the rights to the use and distribution of certain works of creative expression, including books, videos, movies, music, and computer programs.
2. **Patent:** A patent gives its owner the right to exclude others from making, using, selling, and importing an invention for a limited period of time. The patent rights are granted in exchange for enabling public disclosure of the invention.
3. **Trademark:** A Trademark is a Graphical representation that is used to distinguish the goods and services of one party from those of others. A Trademark may consist of a letter, number, word, phrase, logo, graphic, shape, smell, sound, or combination of these things.
4. **Trade Secrets:** Trade secret describes about the general formula of any product and the key behind any organization's progress. It also includes various firms' different secret formulas for the same products which differ in quality.

#### **Advantages of Intellectual Property Rights:**

The advantages of intellectual property rights are as follows:

- 1) IPR yields exclusive rights to the creators or inventors.
- 2) It encourages individuals to distribute and share information and data instead of keeping it confidential.
- 3) It provides legal defense and offers the creators the incentive of their work.
- 4) It helps in social and financial development.
- 5) It inspires people to create new things without fear of intellectual theft.

#### *Ethical issues in Research*

##### **(1) Unethical experimentation**

##### **(a) Research with animals**

Animals play a significant role in research. They are used in a variety of ways by researchers, such as for testing new pharmaceuticals, as teaching tools for medical students, and as experimental subjects for new surgical procedures. Research with animals is necessary and vital

to biomedical research because animal research is frequently a necessary first step toward research involving new medical treatments and pharmaceuticals intended for human use. In order to prevent the mistreatment of animals the United States government first passed the Animal Welfare Act in 1966 (last revised in 1990).

### **(b) Research with human subjects**

**(i) *Respect for Persons***- Informed Consent: Informed consent exists to ensure that all research involving human subjects allows for voluntary participation by subjects who understand what participation entails. Informed consent means that people approached and asked to participate in a research study must: a) know what they are getting involved with before they commit; b) not be coerced or manipulated in any way to participate; and, c) must consent to participate in the project as a subject.

**(ii) *Respect for Persons***- Privacy and confidentiality: Privacy and confidentiality are very important components for research involving human subjects. People have a right to protect themselves, and information gathered during research participation could harm a person by violating their right to keep information about themselves private. The information gathered from people in biomedical studies has a unique potential to be particularly embarrassing, harmful, or damaging.

### **(2) Data management**

The three issues for data management (ethical and truthful data collection, responsibility of collected data, and data sharing) can be addressed by researchers before and during the establishment of a new research project. Ethical data collection refers to collecting data in a way that does not harm or injure someone. Harm and injury could range from outright physical injury to harmful disclosure of unprotected confidential health information. In comparison, truthful data collection refers to data that, once collected, are not manipulated or altered in any way that might impact or falsely influence results. Assigning and ensuring responsibility for collecting and maintaining data is one of the most important ethical considerations when conducting a research project.

### **(3) Research fraud**

Research fraud is publishing data or conclusions that were not generated by experiments or observations but by invention or data manipulation. There are two kinds in research and scientific publishing:

**(i) *Fabrication***: Making up research data and results, and recording or reporting them.

**(ii) *Falsification***: Manipulating research materials, images, data, equipment, or processes. Falsification includes changing or omitting data or results in such a way that the research is not accurately represented.<sup>4</sup>A person might falsify data to make it fit with the desired end result of a study.

### **(4) Guest authorship /Co-authorship**

All individuals who made significant scientific contributions to the research work should be given the opportunity to be included as co-authors. Other persons who contributed to the study

should be acknowledged, but need not be identified as coauthors. Every co-author should be aware of the content of an article to be submitted, agree to its submission, and share appropriate responsibility for the work. Any individual unable to take appropriate responsibility for the article should not be included as a co-author.

#### **(5) Authorship disputes**

Authorship is the process of deciding whose names belong on a research paper. In many cases, research evolves from collaboration and assistance between experts and colleagues. Some of this assistance will require acknowledgement and some will require joint authorship. Each person listed as an author of an article should have significantly contributed to both the research and writing. In addition, all listed authors must be prepared to accept full responsibility for the content of the research article.

#### **(6) Plagiarism and Self-Plagiarism**

Authors should not use, without attribution, text, concepts, data, figures, or tables from another work published either by others or by themselves. Plagiarism of others' works and self-plagiarism are serious breaches of ethics and are not tolerated. If a direct quotation is appropriate, the original source should be properly cited. Figures, tables, and other images reproduced from another source normally require the publisher's permission.

#### **(7) Conflict Of Interest**

Any potential conflicts of interest (e.g., employment, stock ownership, patent licenses, etc.) should be reported to the editorial office. These include personal, academic, political, financial, and commercial gains.

#### **(8) Duplicate/Multiple/Simultaneous Submission**

Duplicate or multiple submissions is the most common ethics violation encountered. It is unethical for authors to publish articles describing essentially the same research result in more than one journal. It is also unacceptable for authors to submit the same manuscript concurrently to more than one journal.

#### **(9) Peer Review**

The two most important ethical concepts in the peer review process are confidentiality and the protection of intellectual property. Reviewers should not know the author (or authors) they are reviewing, and the author (or authors) should not be told the names of the reviewers. Only by maintaining strict confidentiality guidelines can the peer review process be truly open and beneficial. Likewise, no person involved in the peer review process – either the editor, reviewers, or other journal staff – can publicly disclose the information in the article or use the information in a submitted article for personal gain.

#### **(10) Duplicate Publication**

Duplicate publication, or self-plagiarism, is defined as “the publication of an article that is identical or overlaps substantially with an article already published elsewhere, with or without acknowledgment.

### **(11) Redundant Publication**

The redundant or repetitive publication is defined as “the publication of copyrighted material with additional new or unpublished data”.

### **(12) Salami Publication/Salami Slicing**

The “slicing” of research that would form one meaningful paper into several different papers is called “salami publication” or “salami slicing”. Unlike duplicate publication, which involves reporting the exact same data in two or more publications, salami slicing involves breaking up or segmenting a large study into two or more publications. These segments are referred to as “slices” of a study. As a general rule, as long as the “slices” of a broken-up study share the same hypotheses, population, and methods, this is not acceptable practice. The same “slice” should never be published more than once.

### *Different Ethical Issues in Media*

#### **(a) Conflict of Interests**

Conflict of interest occurs when a media person’s obligations or interests clash with his/her duty as a journalist. This practice can take on many forms; let us look at some of them. Many media organisations in India have stakes in other business interests and this interferes with their ability to report objectively on topics that could be detrimental to the said businesses. In such cases, media owners try to tell the stories in a manner that is beneficial to them. They would not want to report and can forbid their editors to report on any issue which could harm their business or affect them adversely. For example, a conflict or clash of interest could crop up while reporting a political story negatively whence the news media owner may be positively inclined to the issue reported. A case in point is that of a large Kolkata business house’s interest in a newspaper.

The owners of a newspaper simultaneously had huge interest in rayon and jute manufacturing businesses in Kerala and West Bengal, respectively. In such situations, particular news media outlets that are owned by these industries, support and propagate the interests of their enterprises by extensive reporting of government policies concerning these industries. It is a reality of the day that almost all news media organisations survive on the strength of advertising revenue. There are several advertisers – multinational corporations and domestic – who dole out heavy and regular ads to the popular news media outlets to seek their editorial support and favourable comment about their business interests. In certain cases, news media outlets have fixated bias against certain parties / leaders /ideologies due to reasons of their own. The viewpoints purveyed and supported by different channels such as NDTV network, The Times Television, Zee network, Doordarshan, or Eenadu television would obviously be poles apart. The reporters of the respective news media would give a core view, but the crux of the story may be tilted to suit the particular news media’s political policy directions.

If a newspaper owner or editor happens to be a nominated legislative member of a party, it would be natural for that media outlet to favour that party, or support its ideology. In such cases, the reporting tilts in their or their party’s favour, which is bound to be apparent in their newspapers’ editorial policies and routine reporting. A reporter of the news media outlet may earn the wrath of his employer or editor by doing a news story that may conflict with the political, religious, economic, and business interest of the media outlet’s owner. Ethical Concerns in Journalism There could also be some occasions when a reporter’s personal views may conflict with the politician or the party that s/he is reporting on.



Also, the views of the editor / proprietor could be in conflict with the politician or the party. It is quite common that some reporters/ correspondents are innately committed to certain political ideologies or political groups, and these are reflected in their stories. In such circumstances, an independent-minded reporter while writing the news story 'objectively' could hurt his/her own or the political interests of the organisation, or could dilute the spirit of the event or issues s/he may be reflecting on. Thus, conflict of interest constitutes a serious issue plaguing news media today.

#### **(b) Bias and Subjectivity**

News reports are often reported in a particular style, angle, and bias in a manner that has led to the suspicion of the news media's intentions and objects. Apart from the bias due to conflict of interest discussed above, reporters quite often get personally involved in doing their stories about certain noted politicians against whom cases of alleged corruption, murder, disproportionate assets, etc., are in courts. Similarly, there are several criminal-politicians, seasoned criminals, and gangsters about whom media persons get to know intimately, and gauge their interests in various national and international scandals. Such personal knowledge gathered when court hearings are in progress leads them to become biased against them. Media persons who cover and see them in court on a regular basis are able to see through their scandalous careers based on the court proceedings. Hence, their reports are likely to get coloured with their personal knowledge of such people's dishonest deeds leading to bias and subjectivity in reporting.

#### **(c) News Source Confidentiality**

With a view to writing in-depth stories, and analysis, or doing investigative reporting, journalists quite frequently report based on information from confidential sources and sensitive documents. Centred on materials so procured or offered, reporters rush through exclusive articles that create a lot of commotion, public furor and even at times political storm. There have been innumerable cases where reporters have published articles using confidential sources and quoting from secret papers and official government documents that have overthrown regimes, ousted governments, put many chief ministers and VIPs behind bars. Often, serious ethical concerns have been expressed where reporters covered special stories, based on information concerning an individual's personal conduct in private life.

#### **(d) Sting Operations**

Deep and all-pervasive ethical concerns have also been involved in issues relating to sting operations, also known as undercover reporting. For example, former BJP president Bangaru Laxman was shown accepting Rs. 1 Lakh for promising political favours in Tehelka's sting operations. In yet another identical case, a sting operation on a Delhi government girls' school teacher was alleged to be involved in a prostitution racket. The sting operation provided to be a fraudulent act; the sting operation was later held and criminal proceedings initiated against him. In the entire process an innocent woman was dragged in a false allegation and the reporter's fraudulent reporting was widely condemned. Due to the news media reporter's penchant for scoops and exclusive stories using sting operations, they get into innumerable traps. While some do triumph with glory and win awards; there are several others who err and get dropped by the wayside and become victims of many unethical allures for the sake of outpacing and beating their rivals, contemporaries and competitors.

#### **(e) Paid News**

According to the Press Council of India (PCI) paid news is ‘any news or analysis appearing in print or electronic media for consideration in cash or kind’. The trend of paid news is on the rise in India, and several print and broadcast media organizations publish favourable news stories in exchange for money. These news stories could be sponsored by organizations, celebrities, high end businessmen, political parties and politicians. A few years ago the PCI asked the government to stop giving advertisements to the newspapers found guilty of paid news.

#### **(f) Intrusion into Privacy**

One of the cardinal principles of news reporting is that news should be treated as a serious business and not considered as entertainment. However, some news media outlets, particularly television channels, often put out news events in a trivially light-hearted style creating derisive impression about the mass media as an institution. Some news media persons in their hunger for exclusive stories about the high and mighty, VIPs, or celebrities, tend to intrude into or peep into the individuals’ totally personal, private and intimate affairs, creating privacy issues for news media outlets as also for themselves. Such stories hurt audiences’ sensitivities and raise questions about their ethical concerns. News media persons should avoid such ventures lest they are caught on the wrong foot. In fact, some Indian language television news channels have been warned by the Ministry of Information & Broadcasting to shun such practices.

#### **(g) PR Press Releases Masquerading as News**

It is a well-known fact that several news media persons end up as public relations personnel or corporate communication specialists due to heavy remunerations offered by many noted multinational companies and corporate firms. These PR persons are often friendly with news media persons and tend to push the press notes and press releases of their companies masquerading as news stories. Such stories are very easily discernible and can hurt the ethical views of news media’s overall conduct and general public’s sensibilities.

Some other ethical concerns pertain to plagiarism, sensationalism, and offensive reporting. Plagiarism of news items from other sources degrades the media’s prestige and lowers the reporter’s position in the profession. And, when the media audiences discover such tendencies in a particular mass media’s practices, they start losing faith and trust in its credibility and integrity in its overall news operations. They may occasionally present certain exclusive stories in a manner that’s deemed outrageous and outright offensive. Sensational reports on rapes, kidnapping, molestation, murders, and many other crimes relating to women not only are in bad taste, but also demeaning for women.

#### *Different Ethical Issues in CSR*

**(a) Environmental Degradation & Green-washing:** Companies may deceptively market products as eco-friendly (“green-washing”) to gain reputation, while causing damage in manufacturing or failing to meet true sustainability standards.

**(b) Labor Practices and Human Rights:** Ethical issues arise concerning inadequate wages, long working hours, poor safety measures, and the exploitation of labor in developing nations.

- (c) **Transparency and Accountability:** A major issue is the lack of transparency in reporting, where companies may misrepresent their social and environmental impact, or fail to accurately disclose financial contributions.
- (d) **Conflicts of Interest:** Ethical dilemmas arise when personal, financial, or familial interests of decision-makers conflict with their professional duty to act in the best interest of stakeholders.
- (e) **Ethical Use of Technology:** Tech companies face dilemmas regarding data privacy, user surveillance, and the potential negative societal impacts of artificial intelligence and digital innovations.
- (f) **Marketing and Product Safety:** Misleading advertising, deceptive sales promotions, and failing to disclose risks related to product safety are significant ethical lapses in the marketing component of CSR.
- (g) **Corporate Governance and Integrity:** Ensuring that corporate actions comply with legal frameworks (e.g., Sarbanes-Oxley) and uphold integrity in financial reporting, avoiding scandals.

#### *Meaning and Significance of Ethical Codes*

Ethical codes are sets of principles, values, and standards of conduct designed to guide professionals and organizations in acting with honesty, integrity, and fairness. They serve as frameworks for decision-making, ensuring compliance with social norms and legal requirements while fostering trust among stakeholders.

#### *Significance of Ethical Codes*

- (a) **Guiding Behavior:** They define acceptable and unacceptable behavior, providing a clear reference for handling complex, gray-area dilemmas.
- (b) **Building Trust:** They foster confidence among customers, employees, and the public by demonstrating a commitment to high moral standards.
- (c) **Risk Management:** They help protect organizations from legal liability, regulatory fines, and reputation damage.
- (d) **Consistency:** They ensure uniform behavior across an organization, aligning individual actions with organizational goals.

#### *The Limitations of Ethical Codes*

- (a) **Vagueness and Generality:** Ethical codes often use broad language, making them difficult to apply directly to nuanced, real-world situations.
- (b) **Inability to Cover All Scenarios:** It is impossible for a code to anticipate every moral problem or technological advancement, leaving gaps in coverage.

